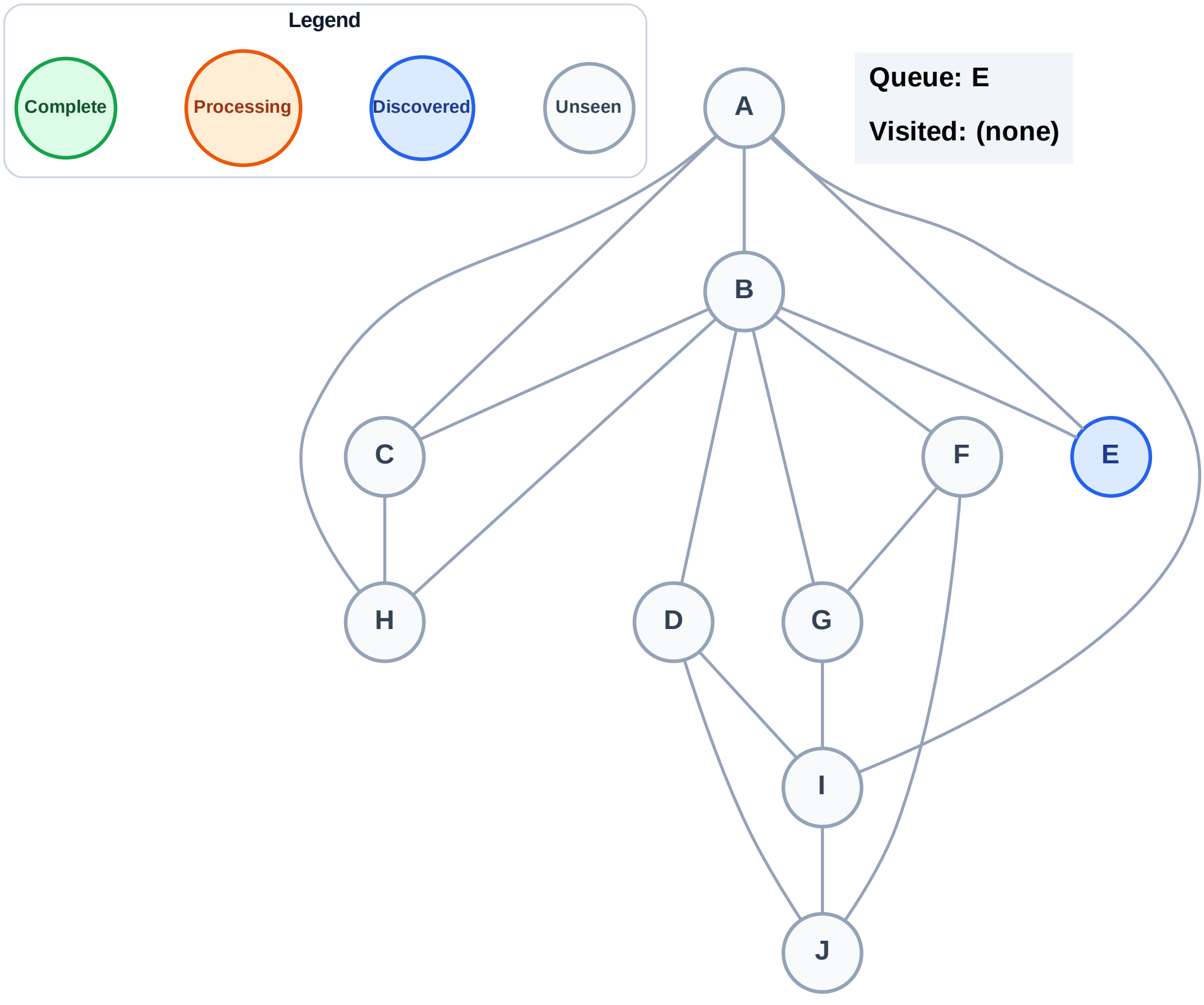
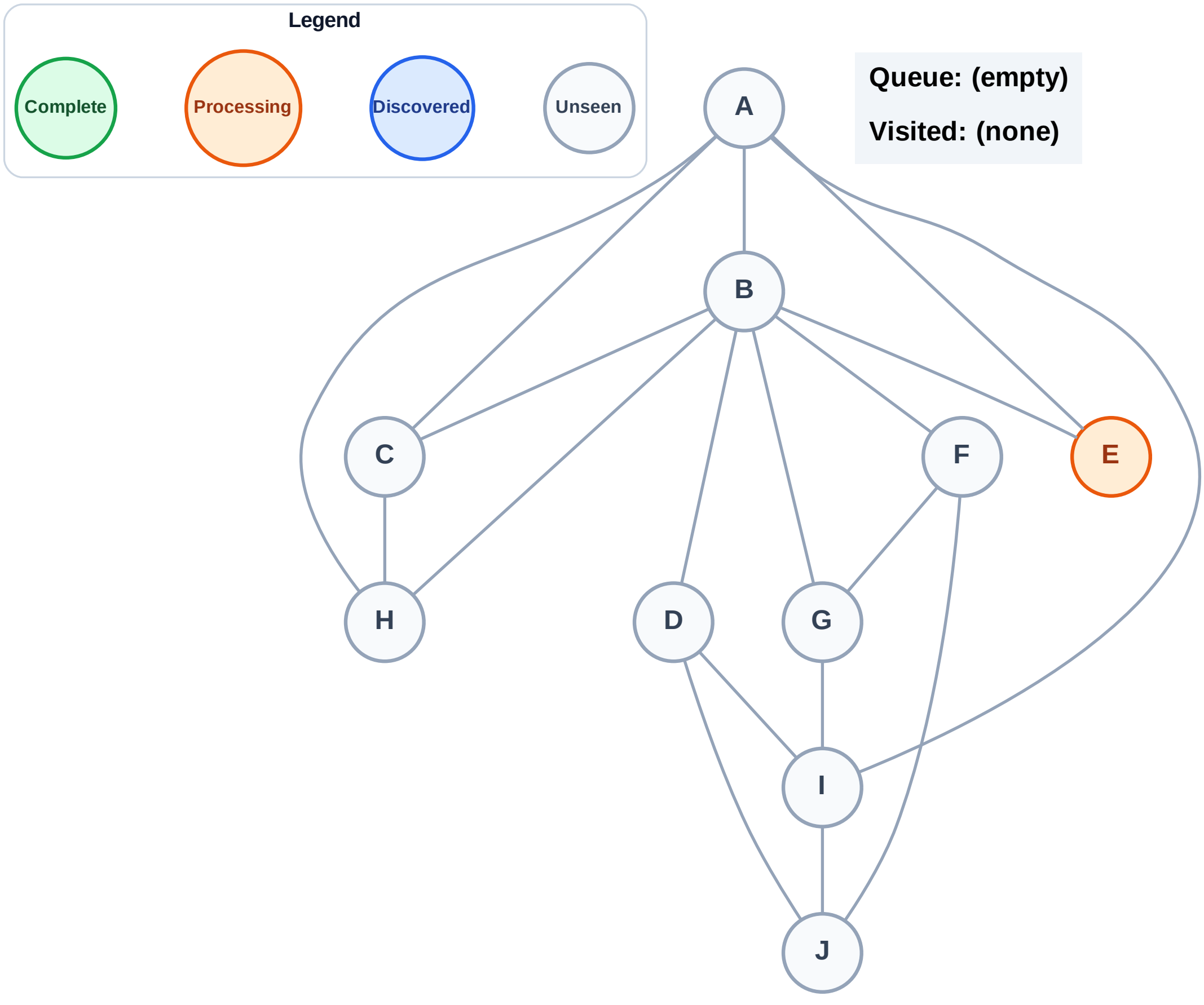
BFS Traversal

Step 1: Start at E



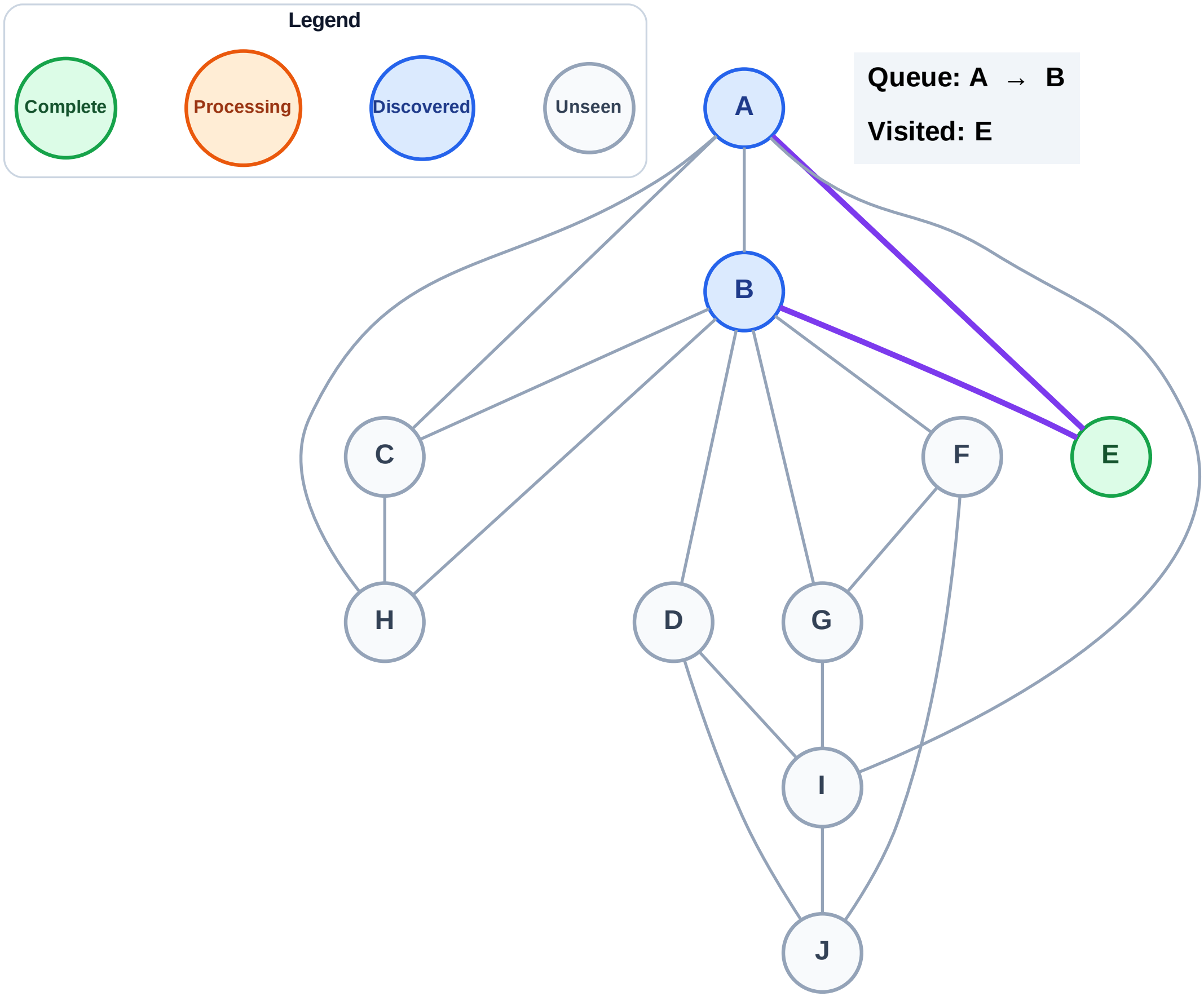
BFS Traversal

Step 2: Process node E



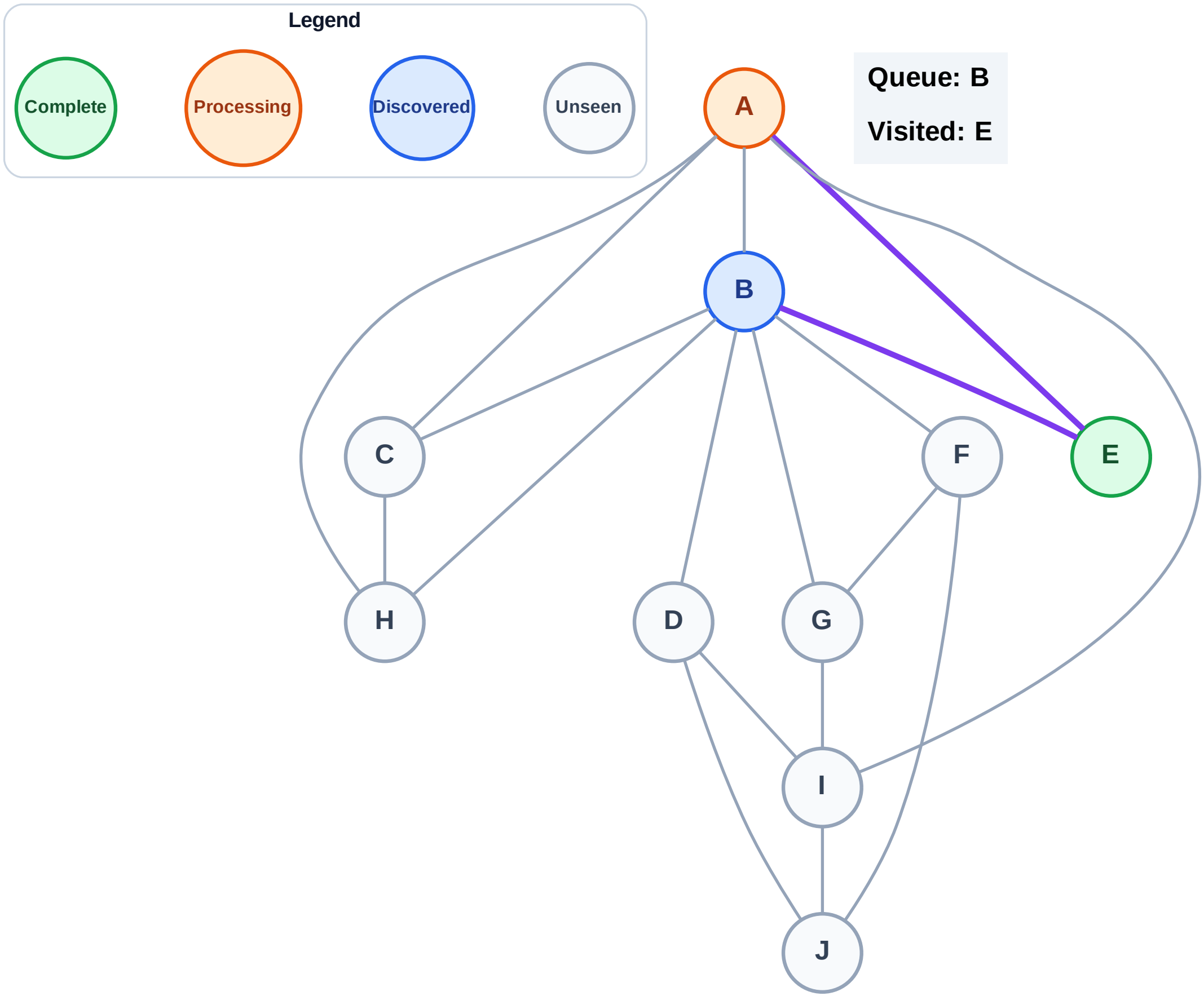
BFS Traversal

Step 3: Discover A, B from E



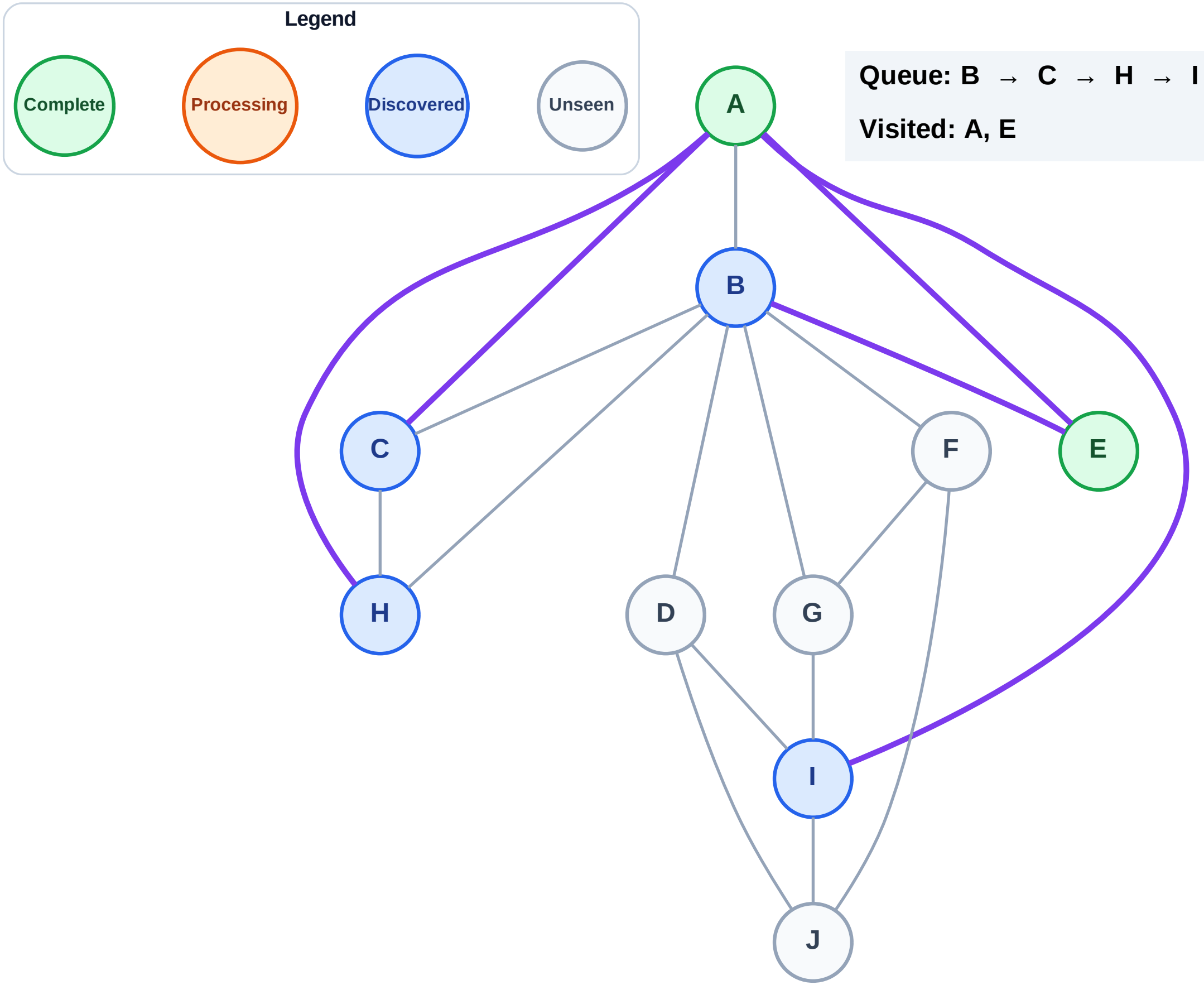
BFS Traversal

Step 4: Process node A



BFS Traversal

Step 5: Discover C, H, I from A

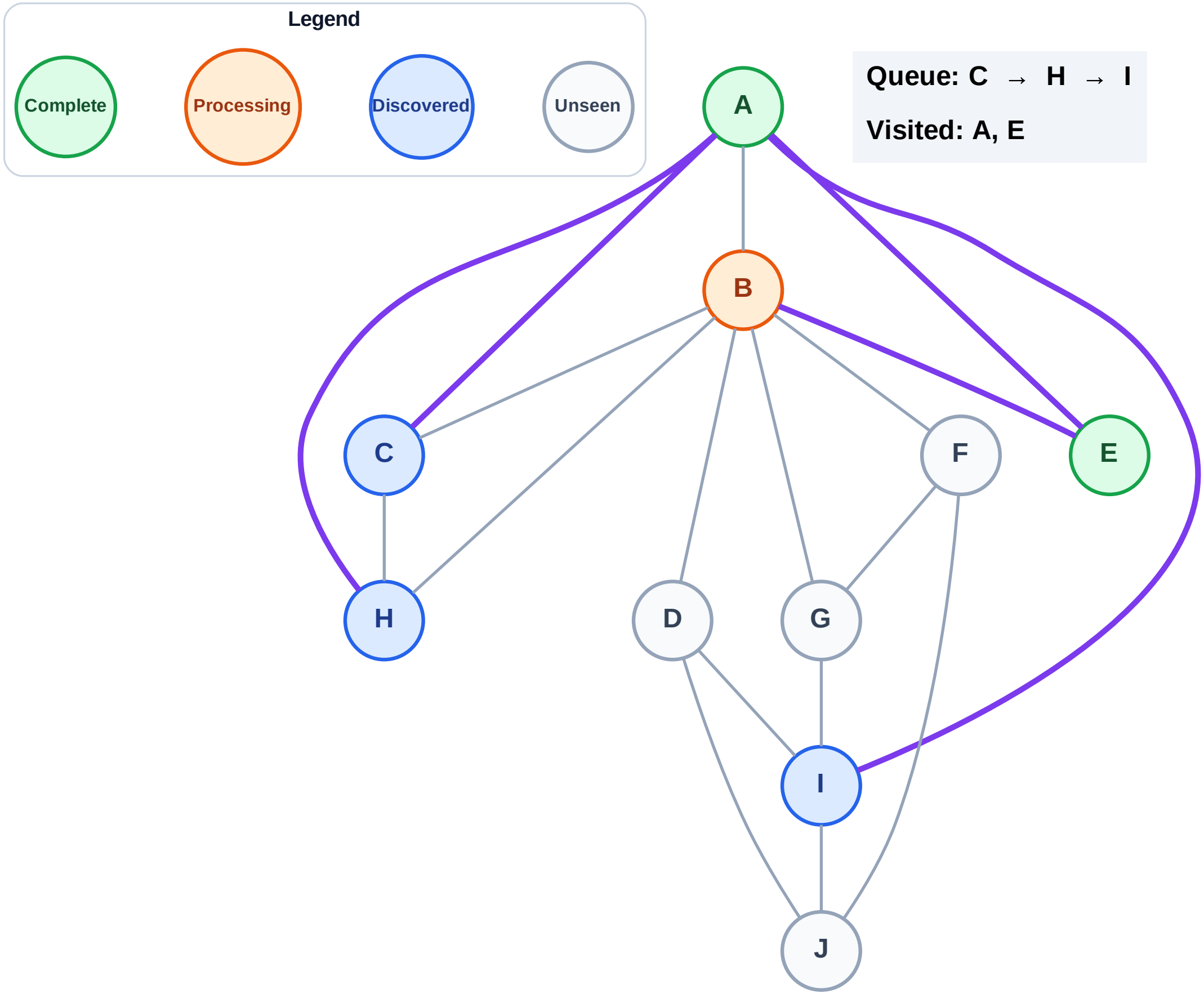


Queue: B → C → H → I

Visited: A, E

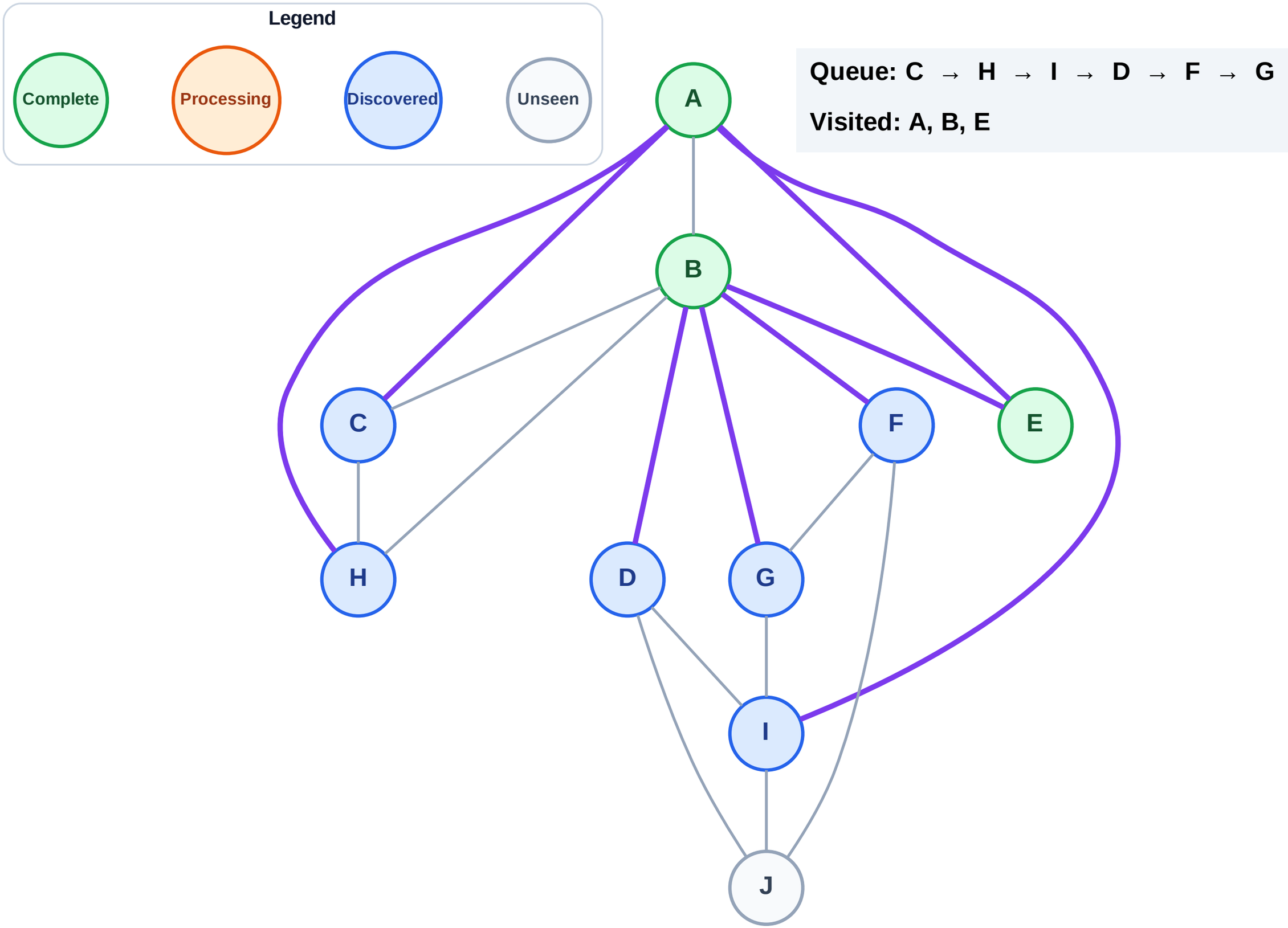
BFS Traversal

Step 6: Process node B



BFS Traversal

Step 7: Discover D, F, G from B

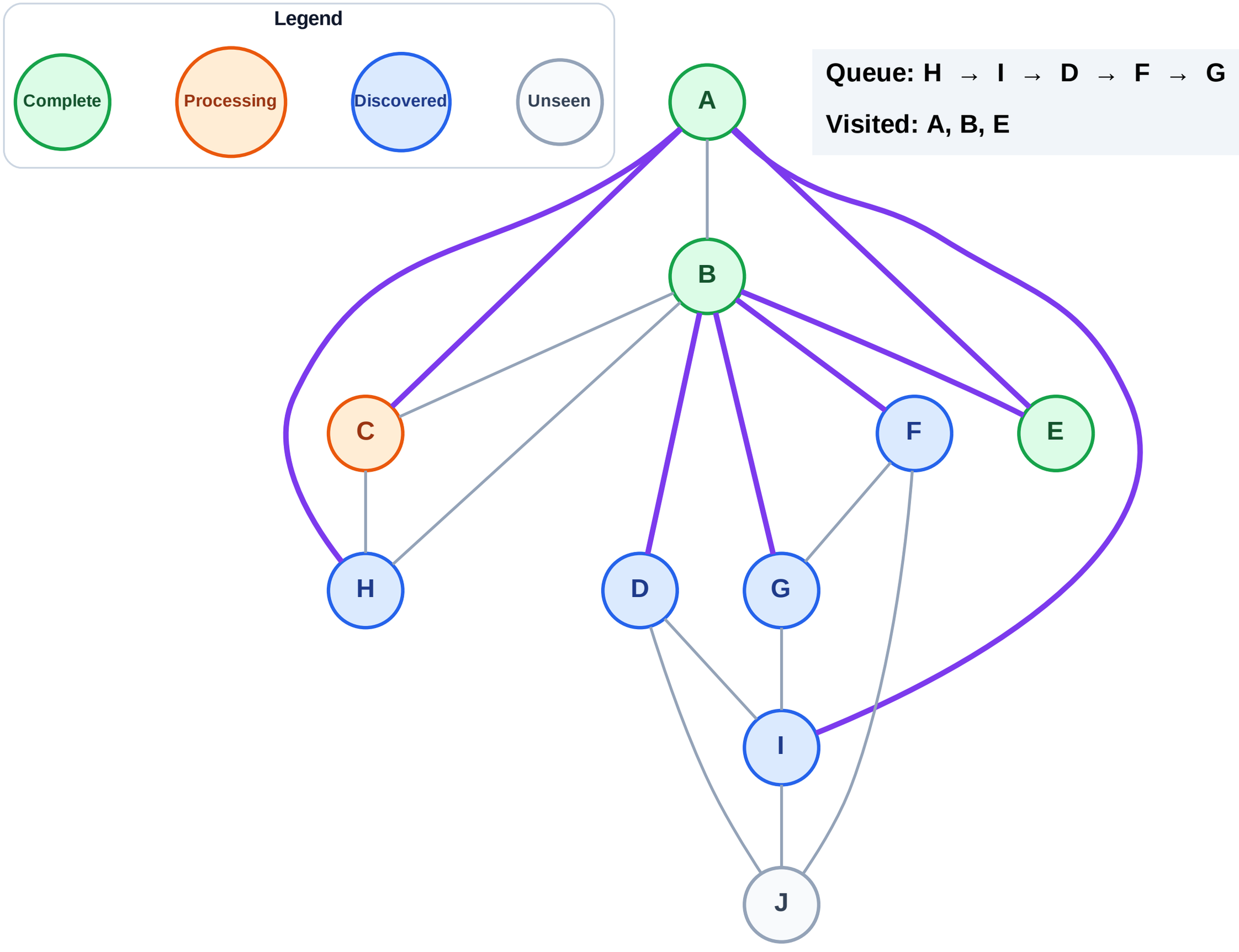


Queue: C → H → I → D → F → G

Visited: A, B, E

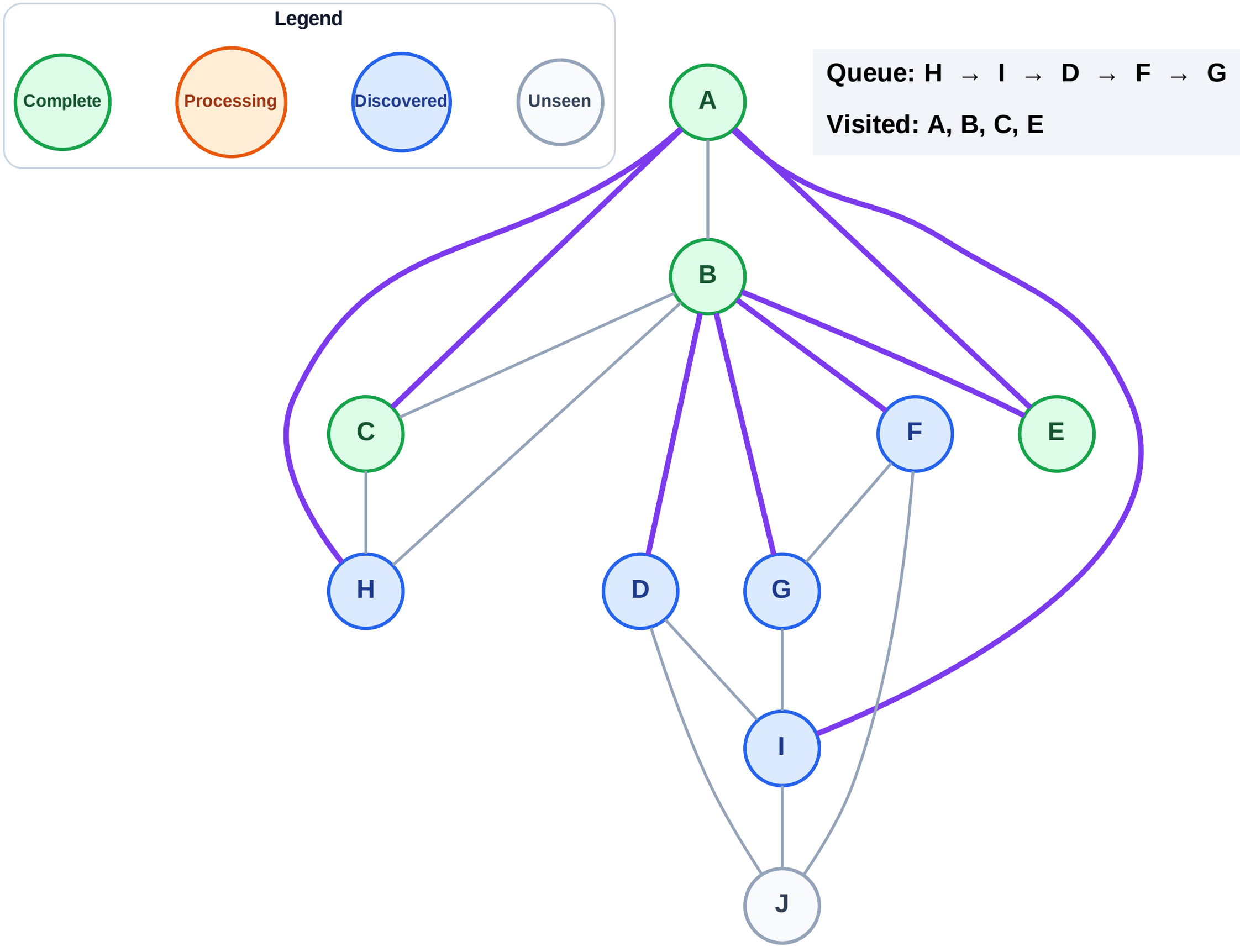
BFS Traversal

Step 8: Process node C



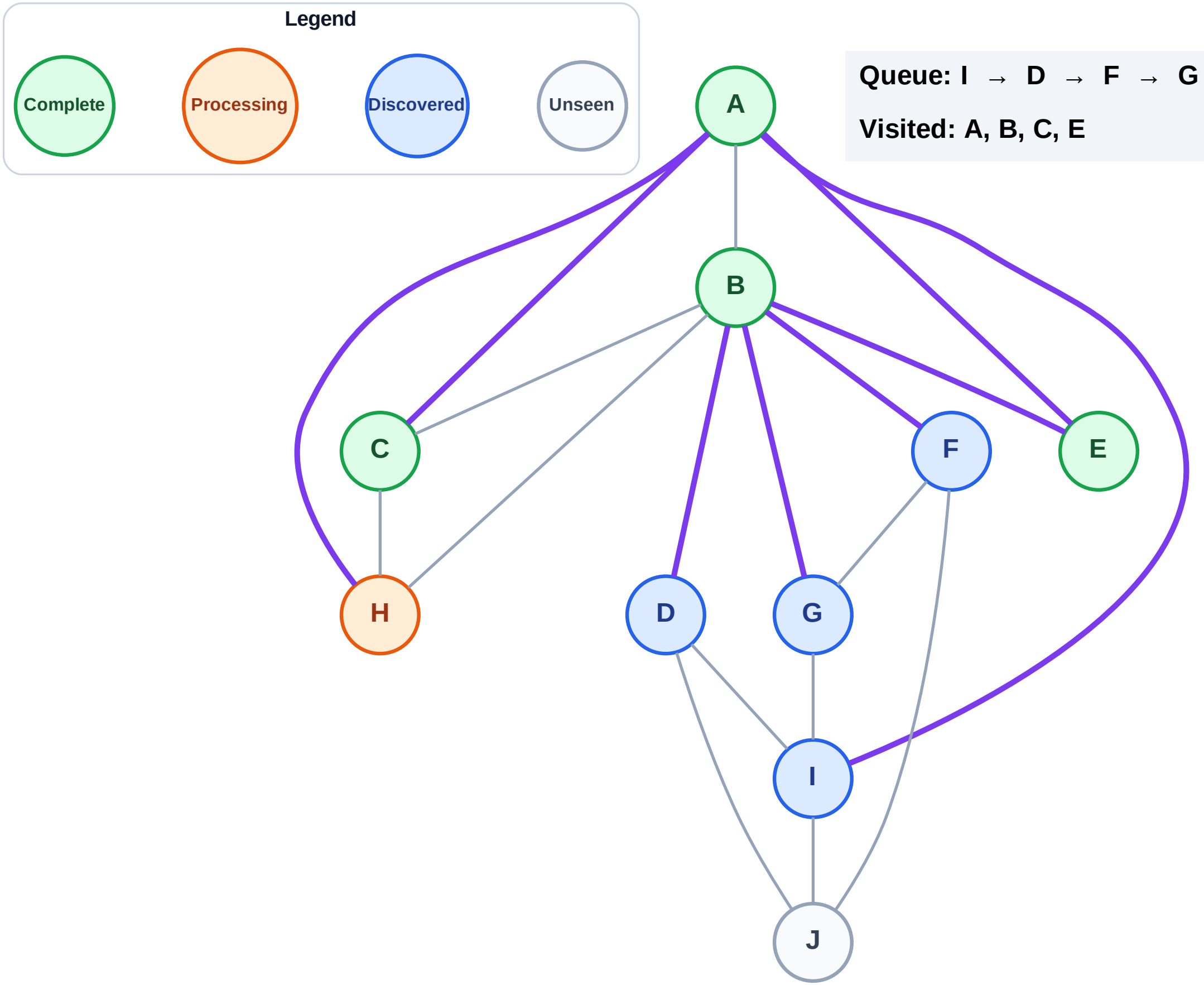
BFS Traversal

Step 9: No new neighbors from C



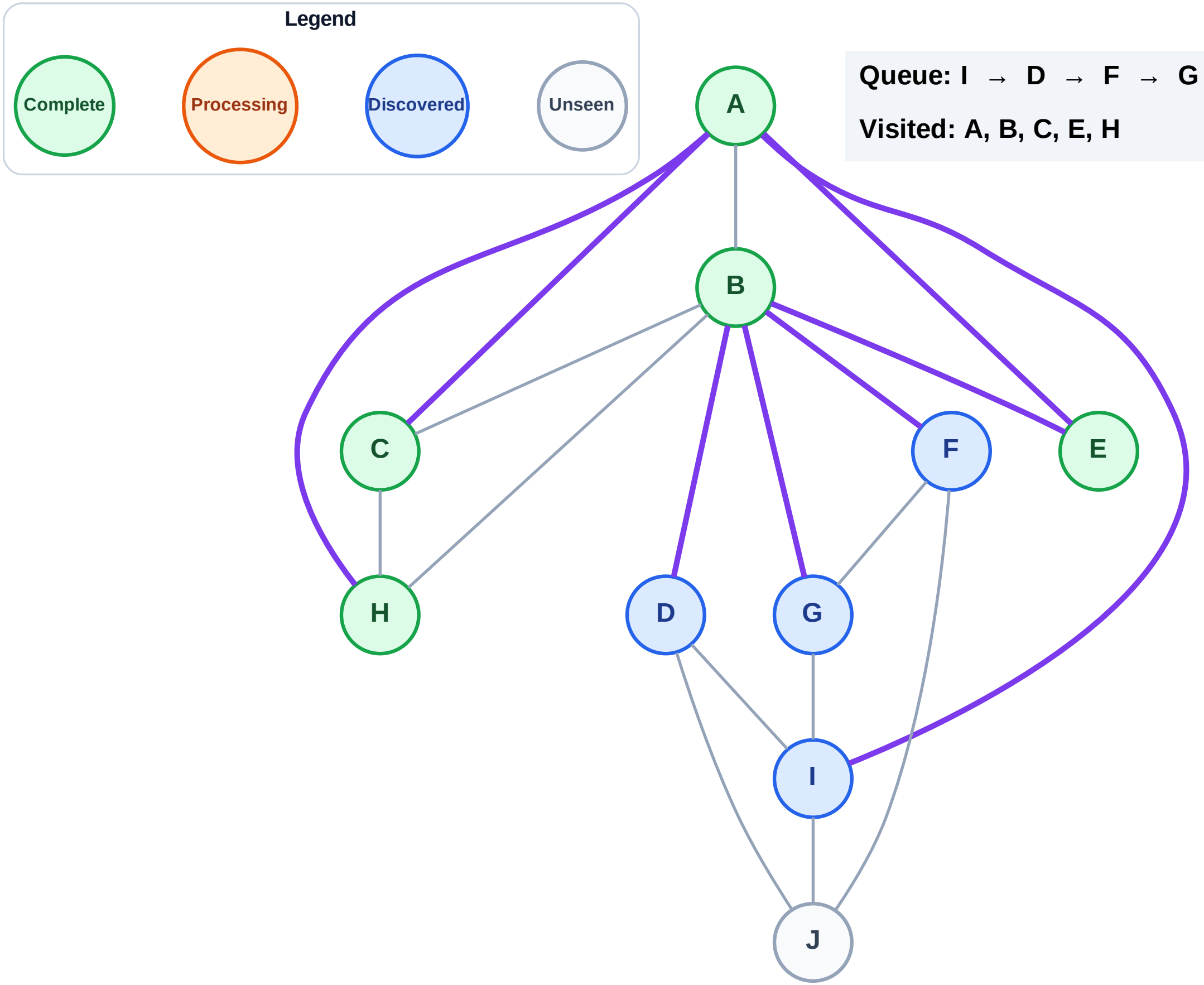
BFS Traversal

Step 10: Process node H



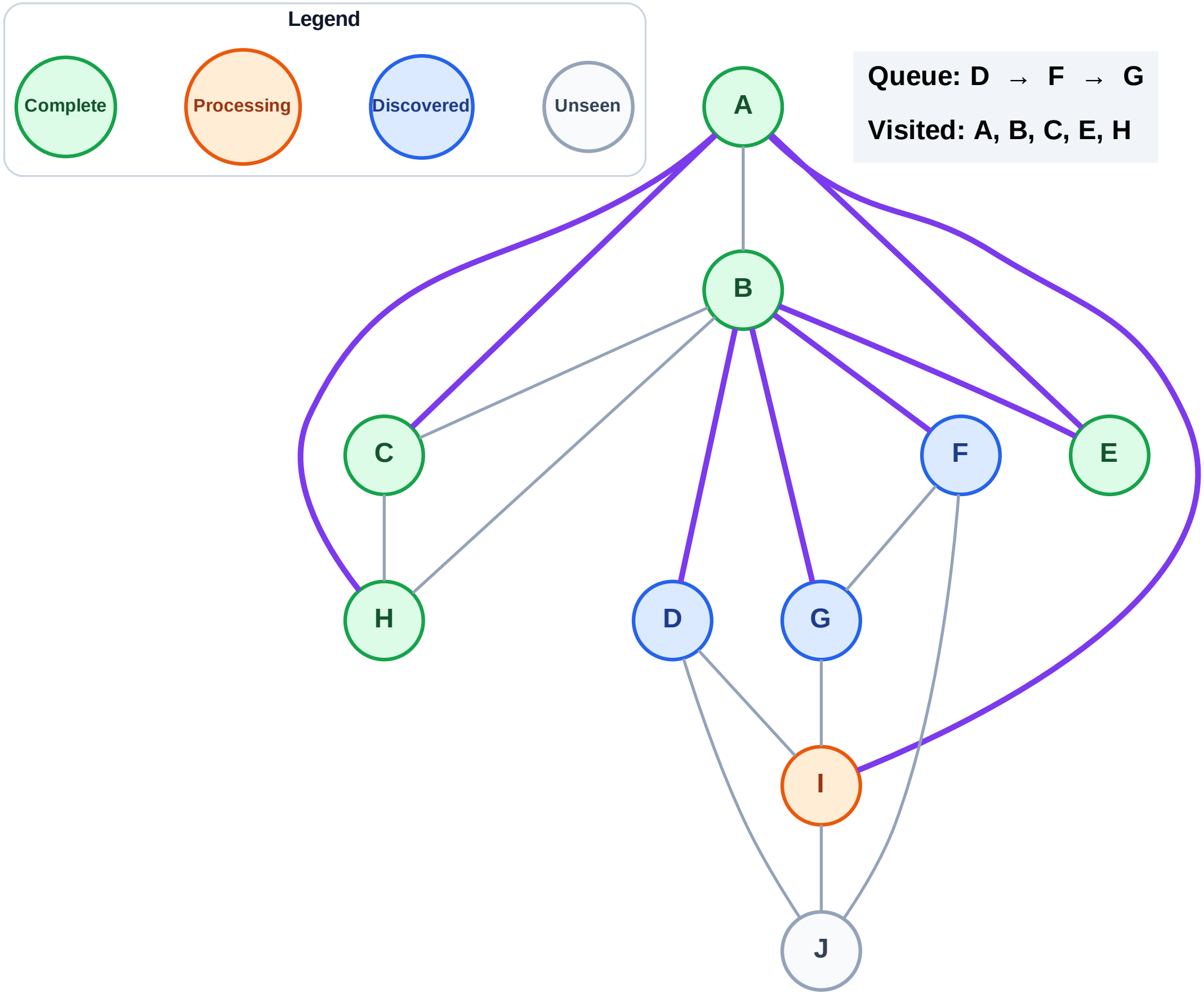
BFS Traversal

Step 11: No new neighbors from H



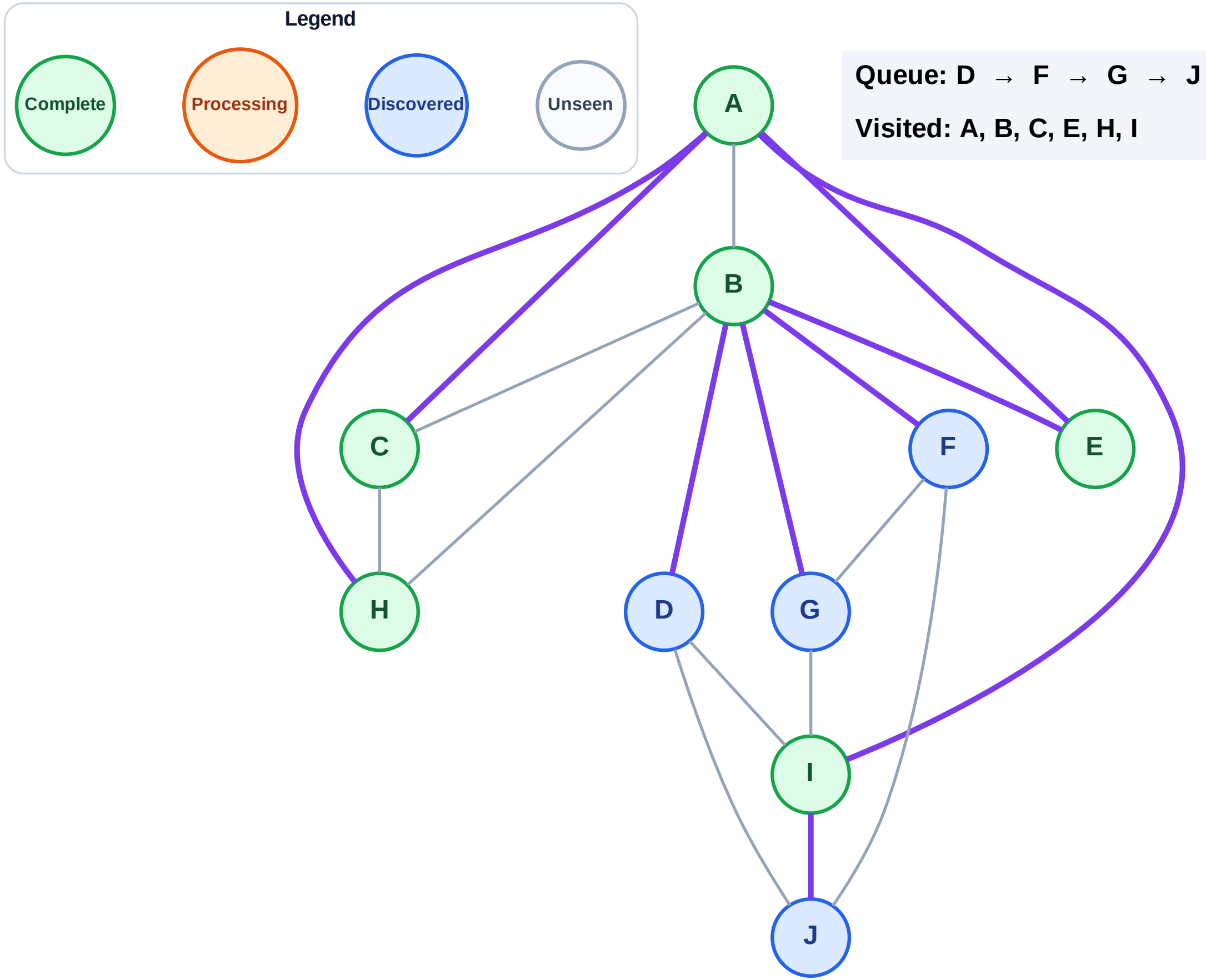
BFS Traversal

Step 12: Process node I



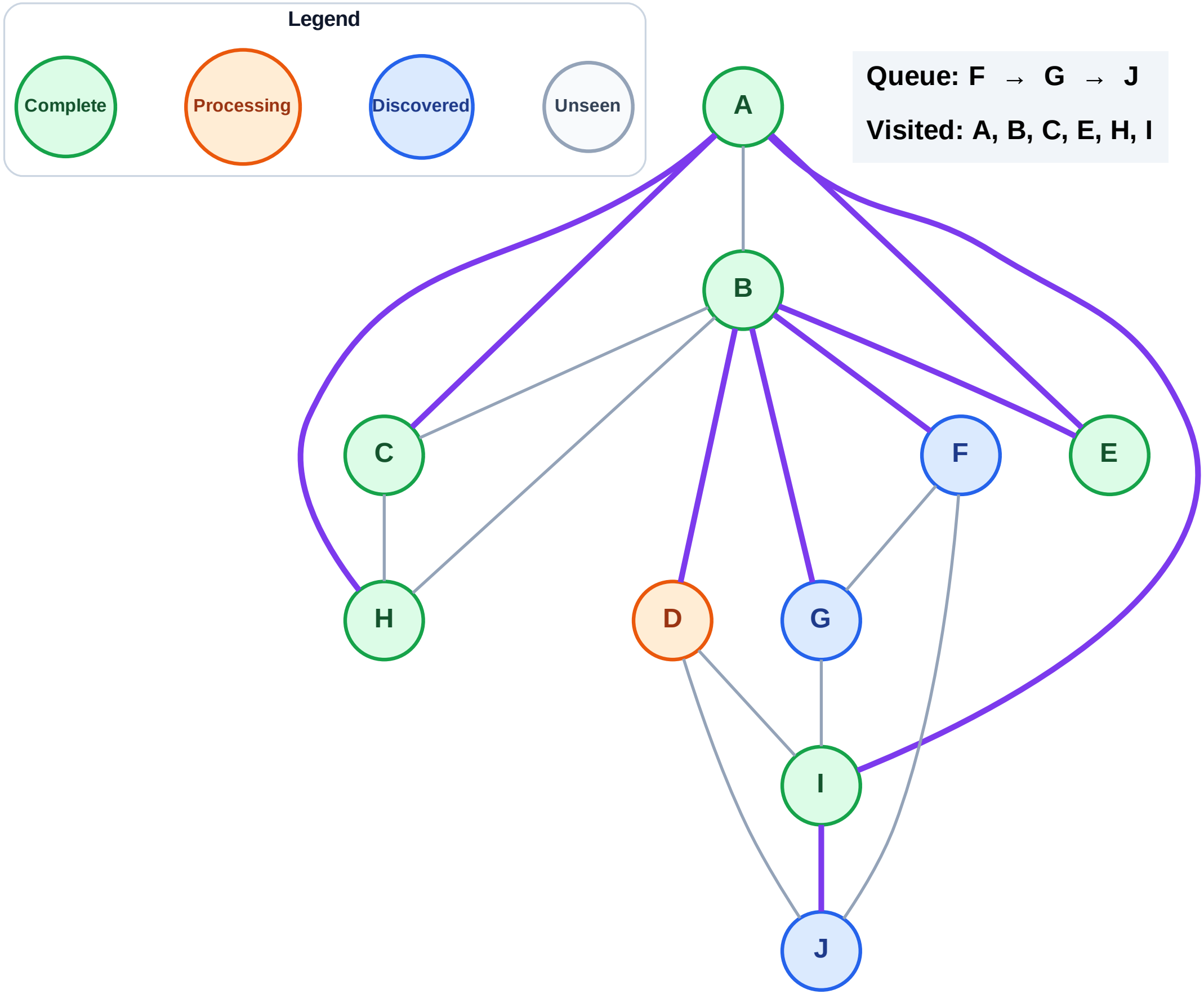
BFS Traversal

Step 13: Discover J from I



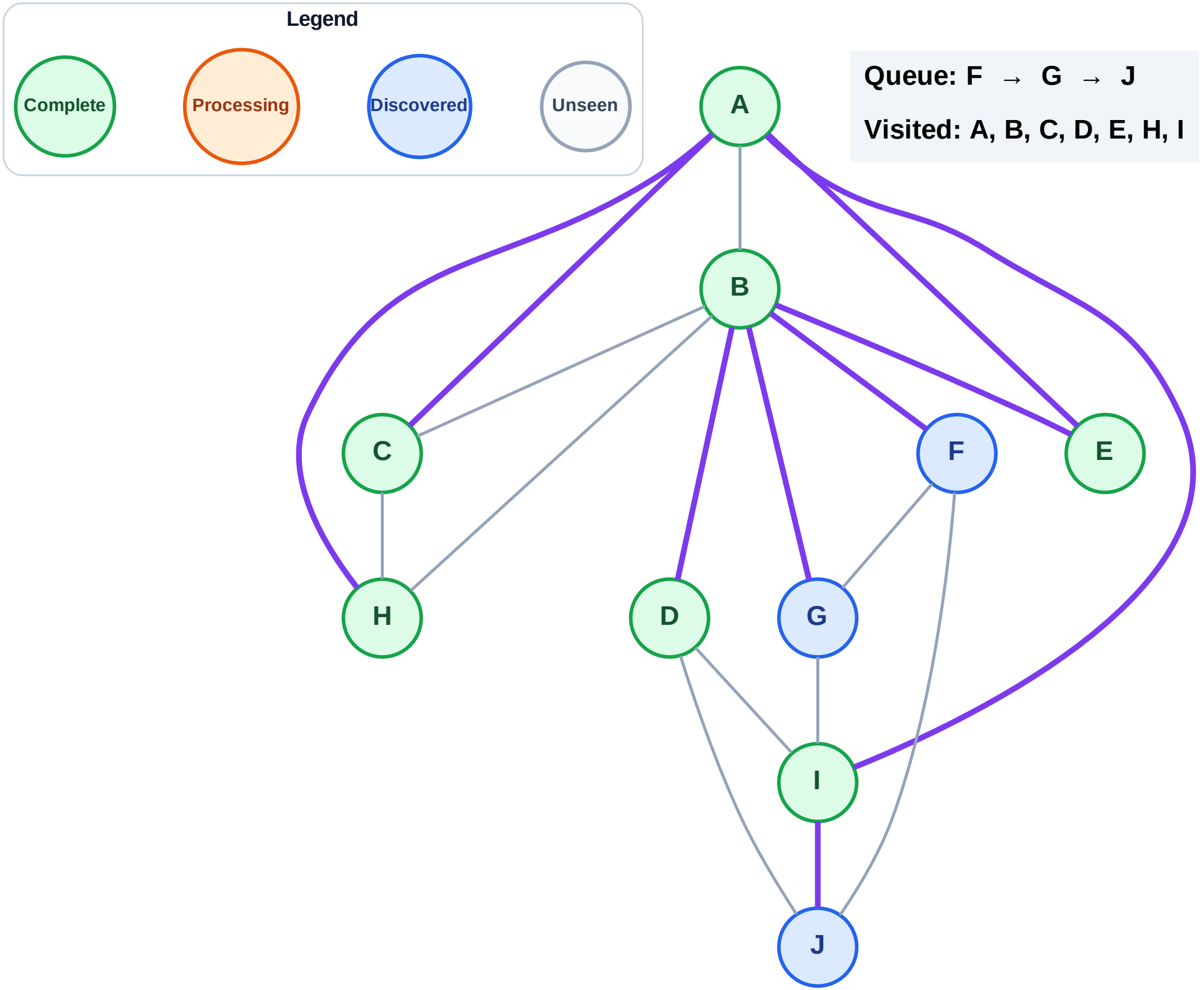
BFS Traversal

Step 14: Process node D



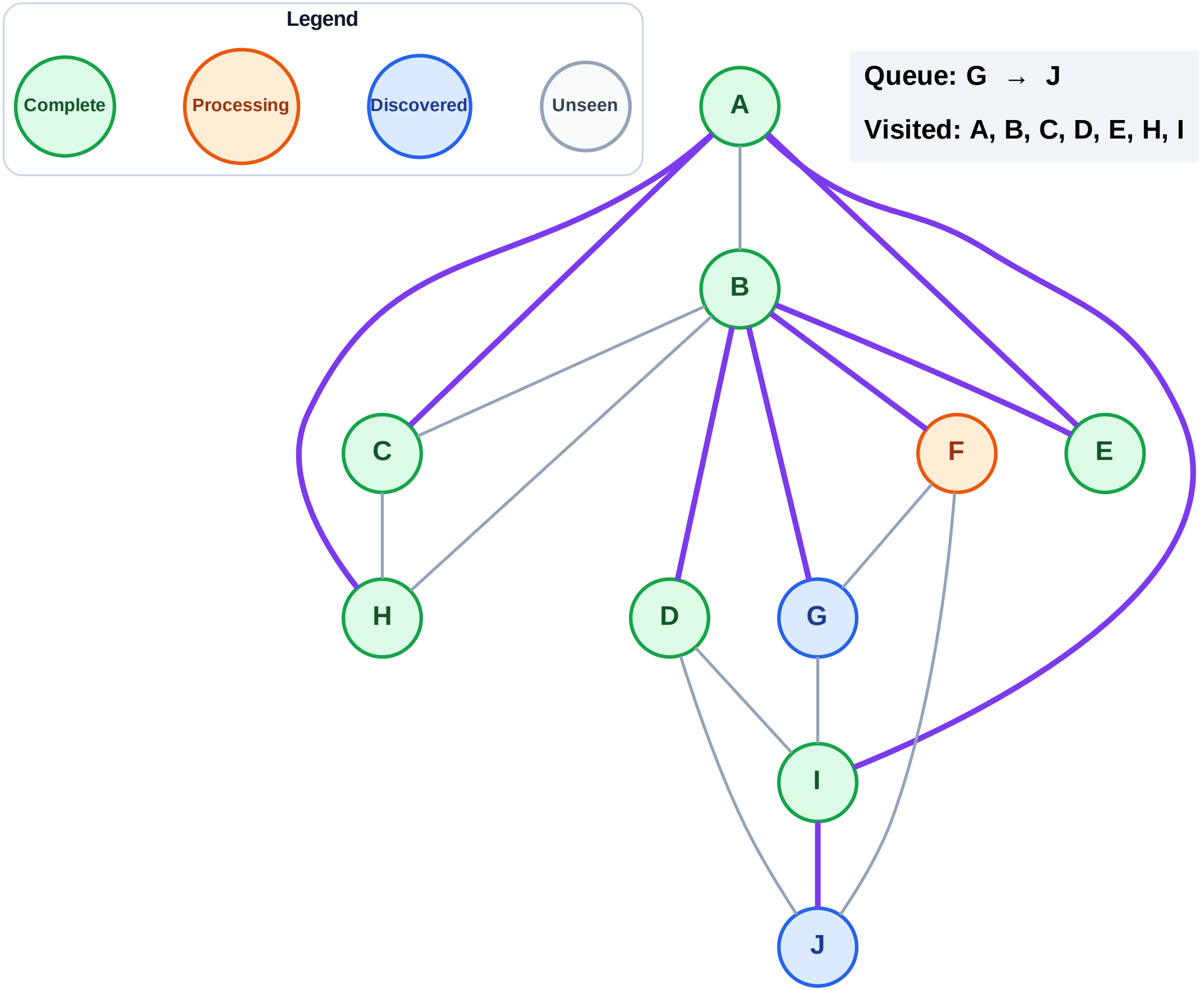
BFS Traversal

Step 15: No new neighbors from D



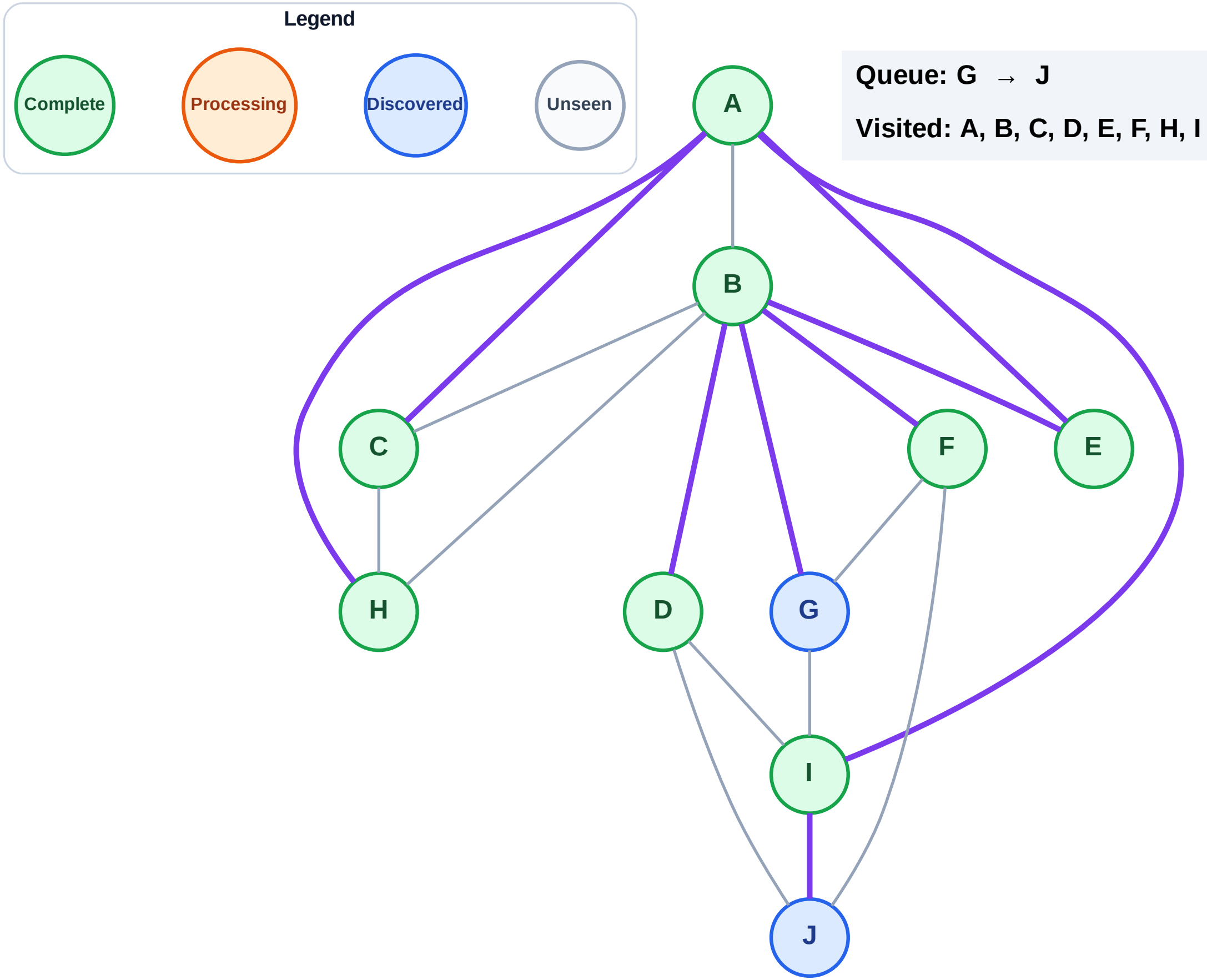
BFS Traversal

Step 16: Process node F



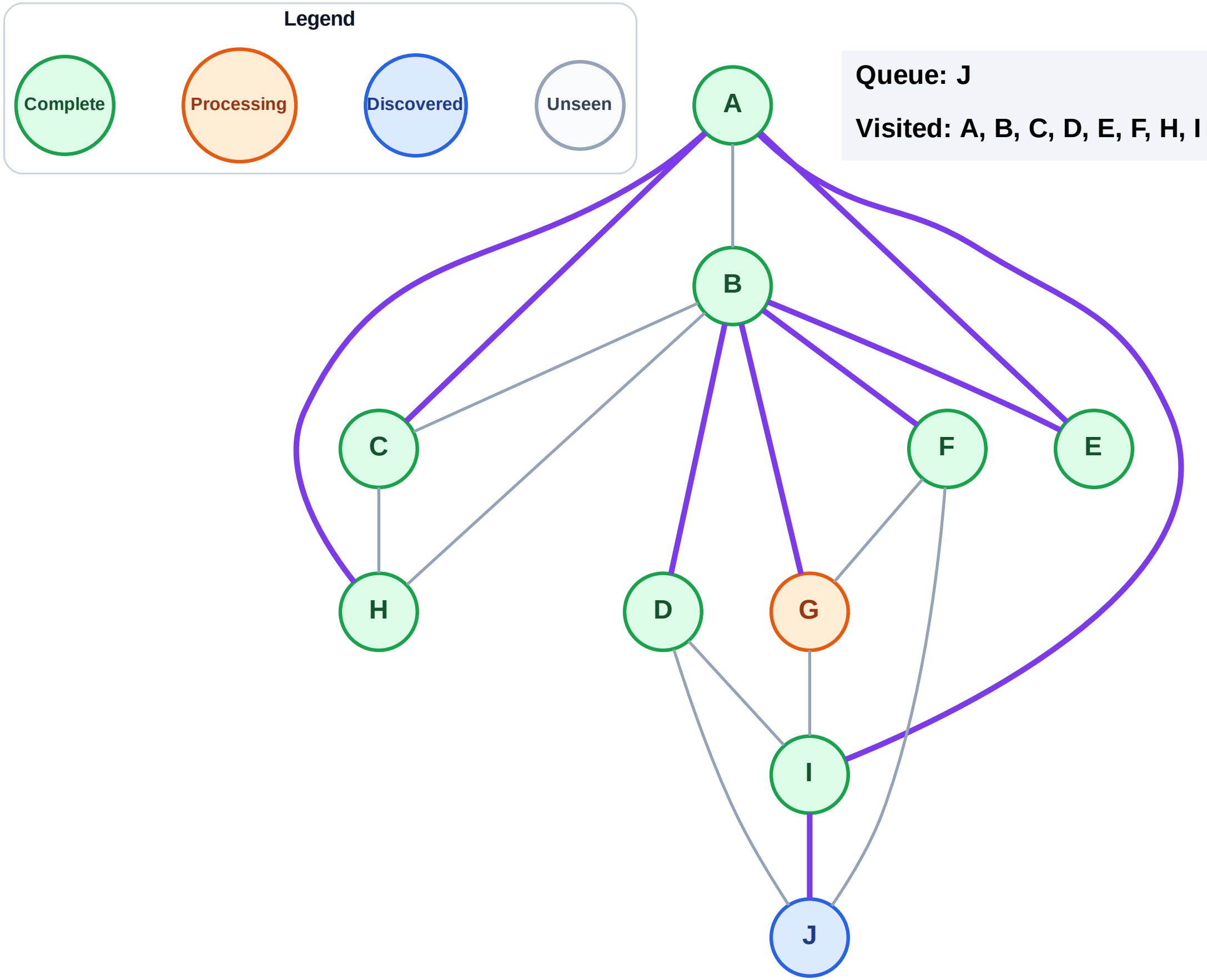
BFS Traversal

Step 17: No new neighbors from F



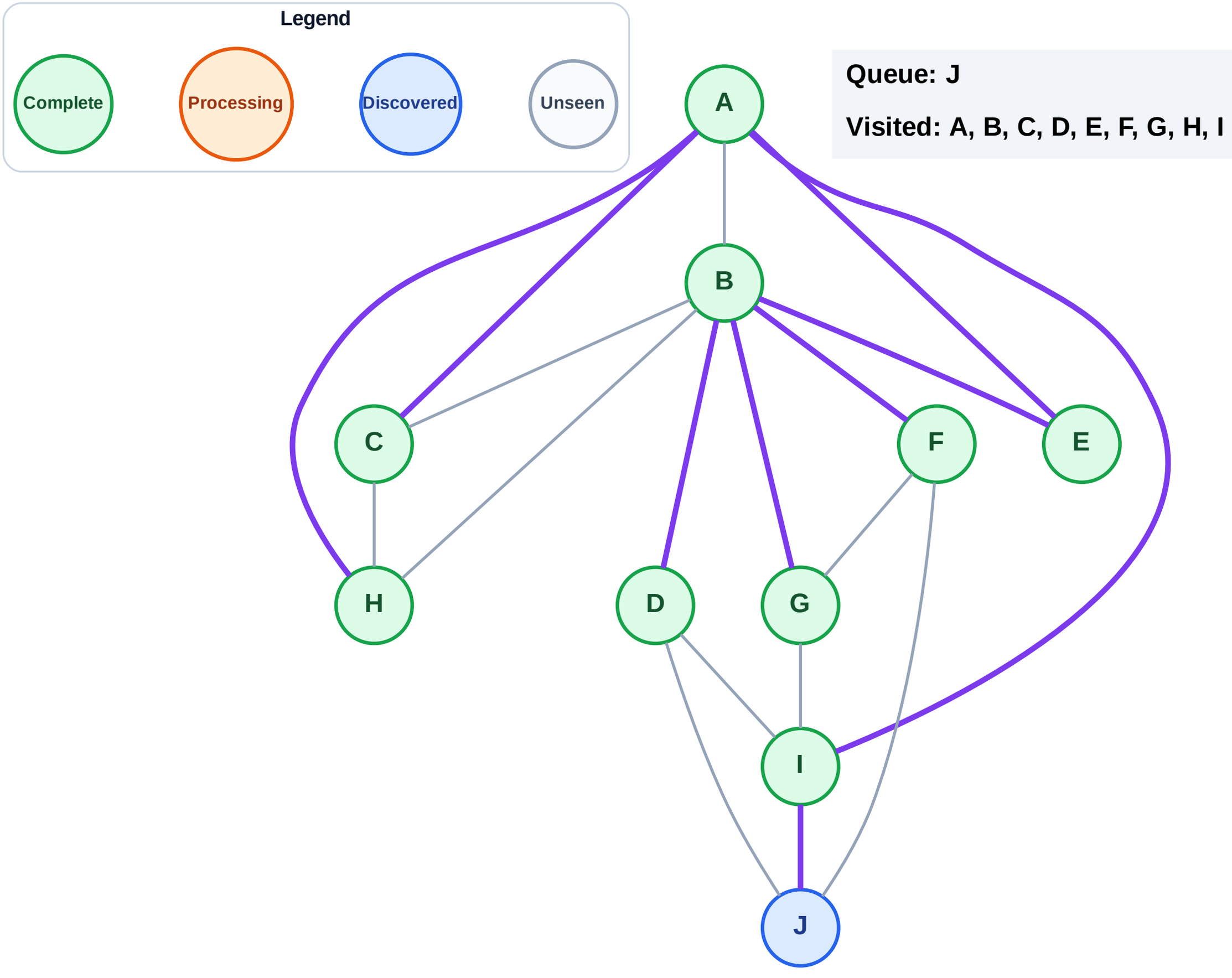
BFS Traversal

Step 18: Process node G



BFS Traversal

Step 19: No new neighbors from G

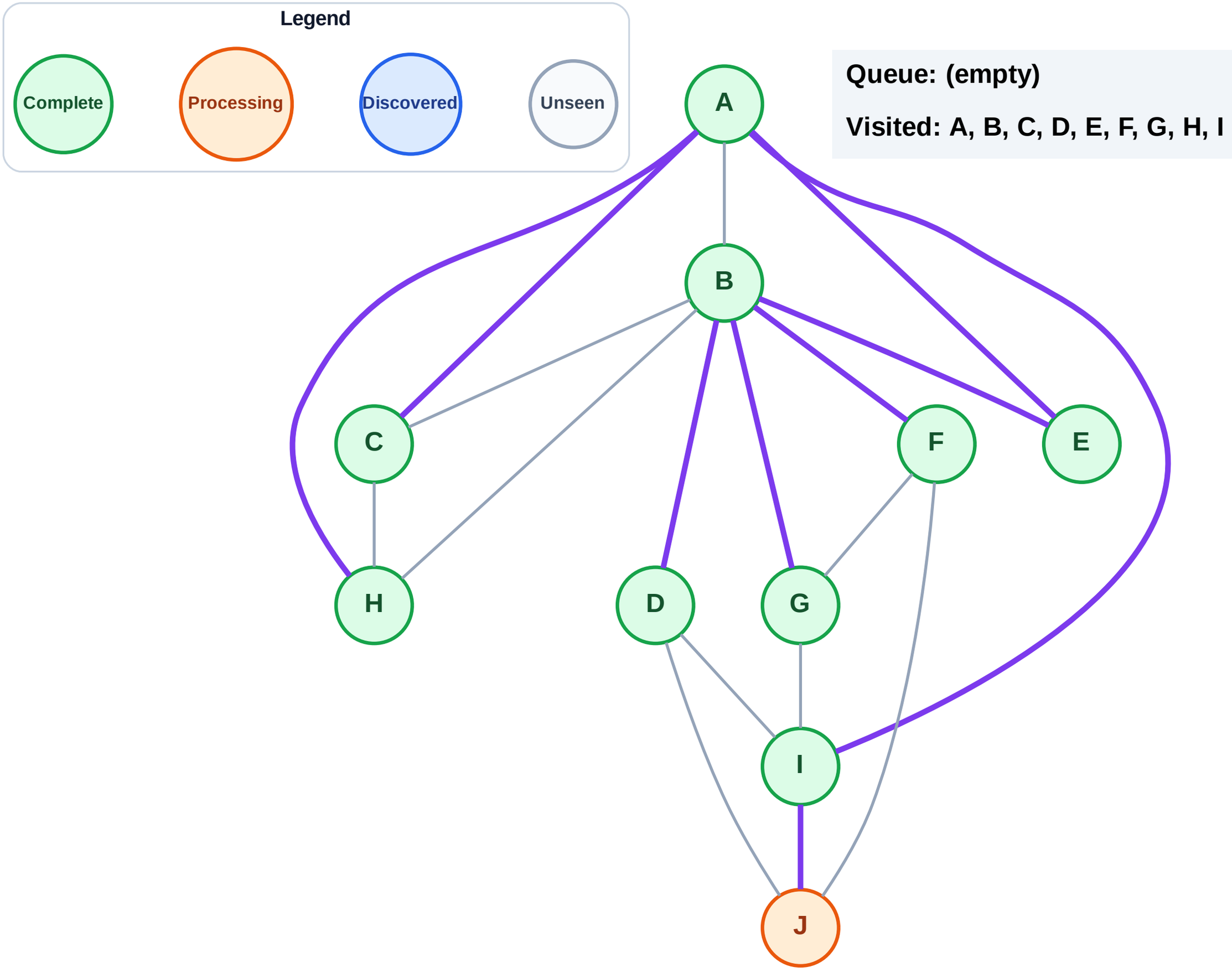


Queue: J

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 20: Process node J

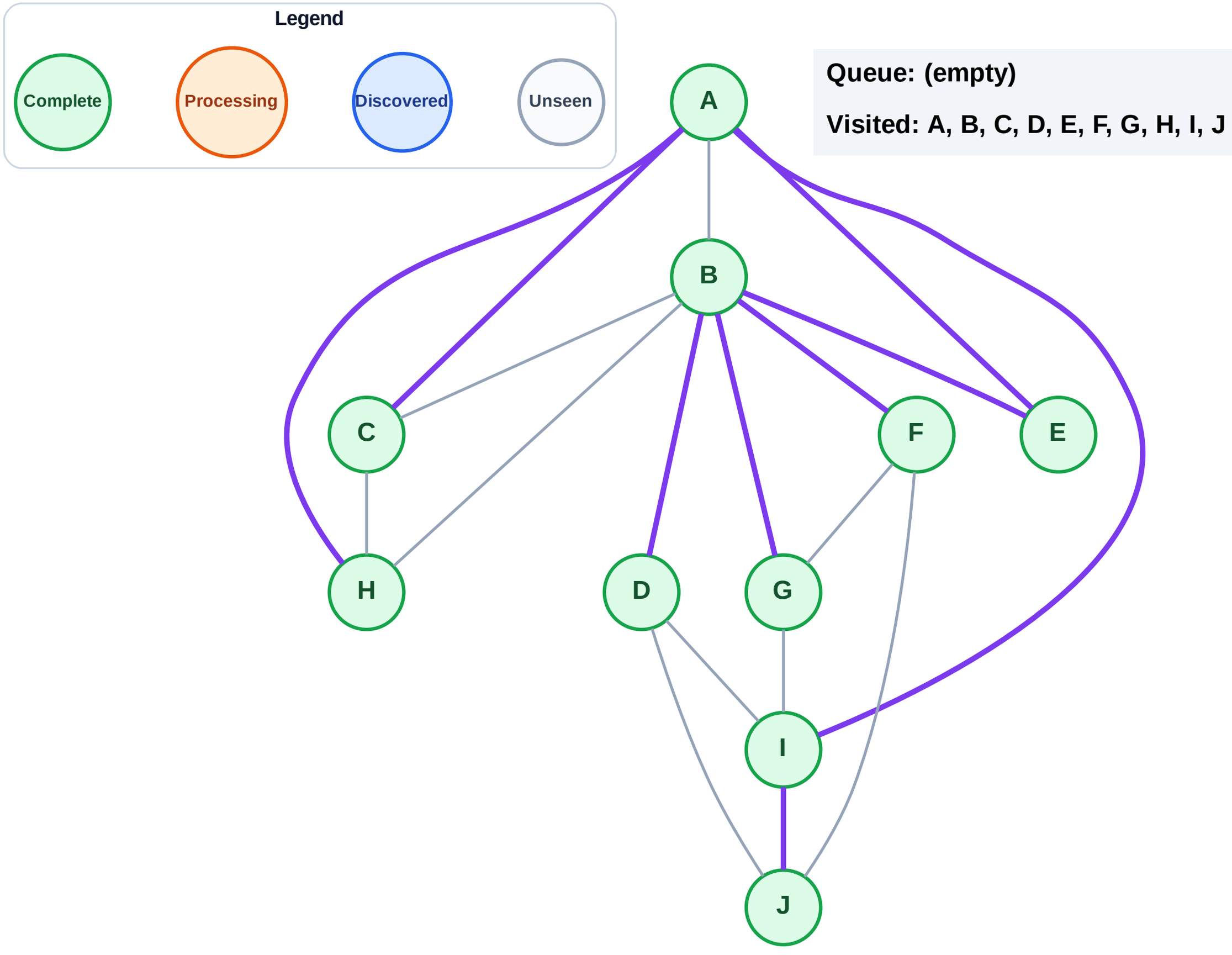


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I

BFS Traversal

Step 21: No new neighbors from J

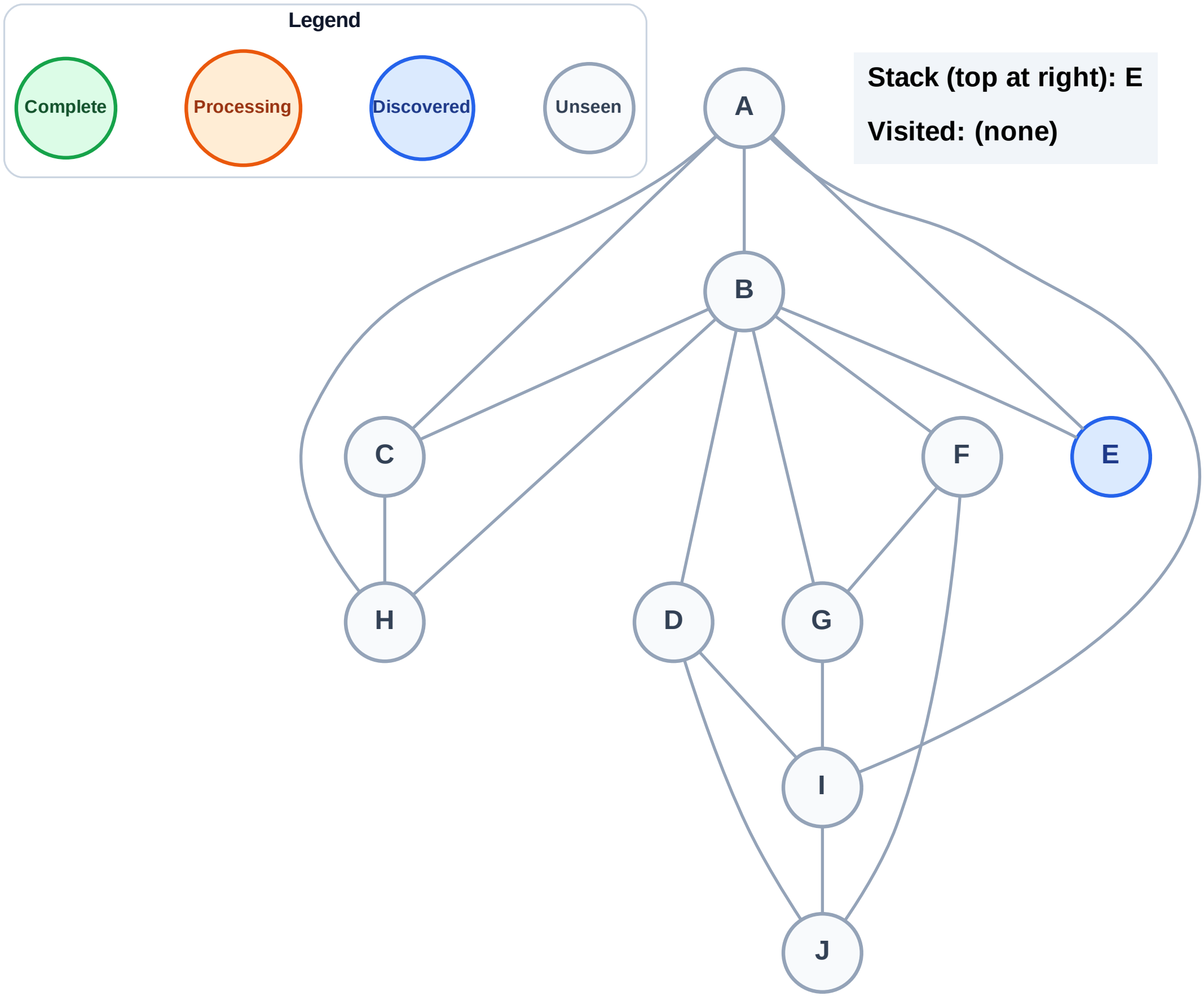


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

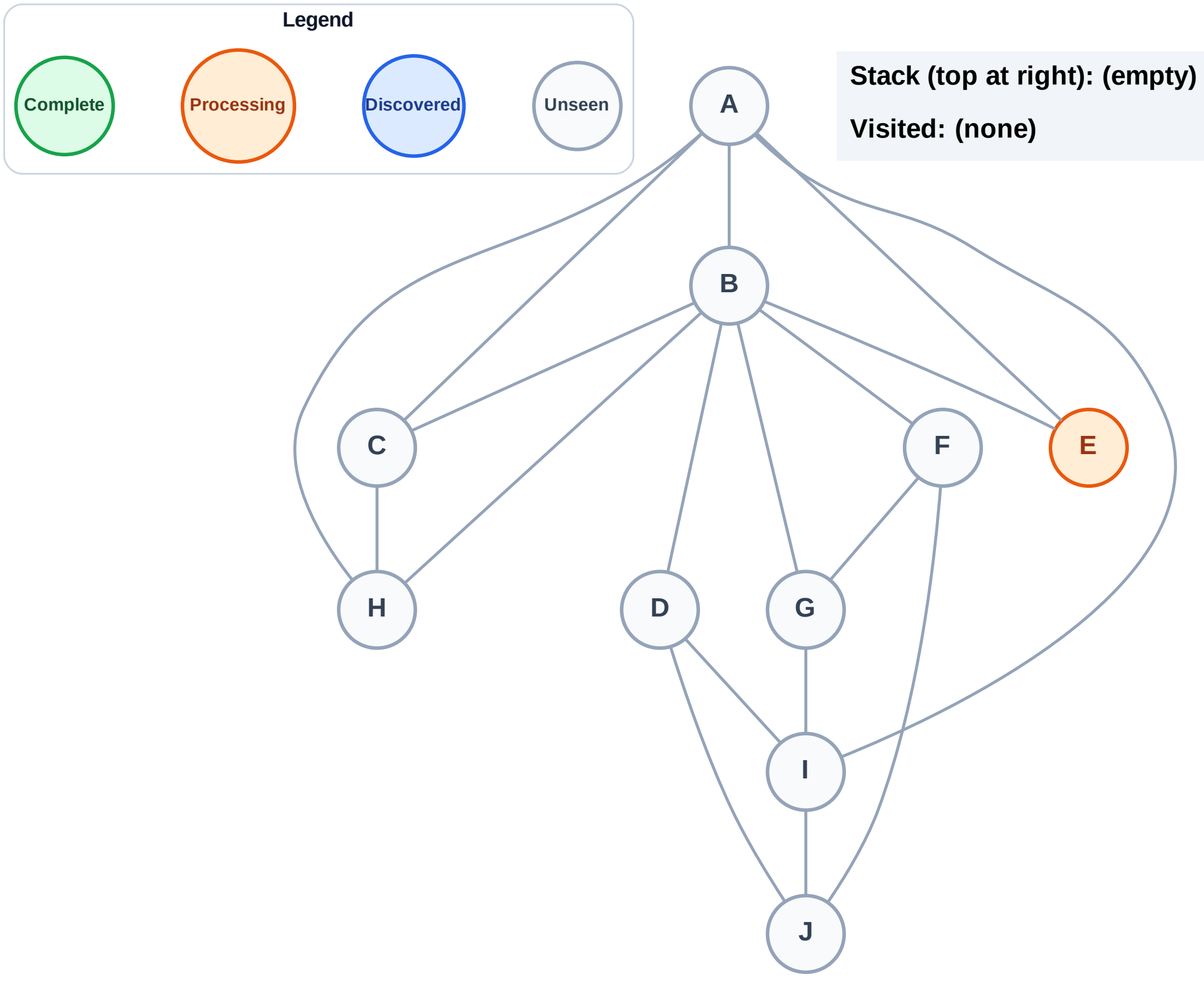
DFS Traversal

Step 1: Start at E



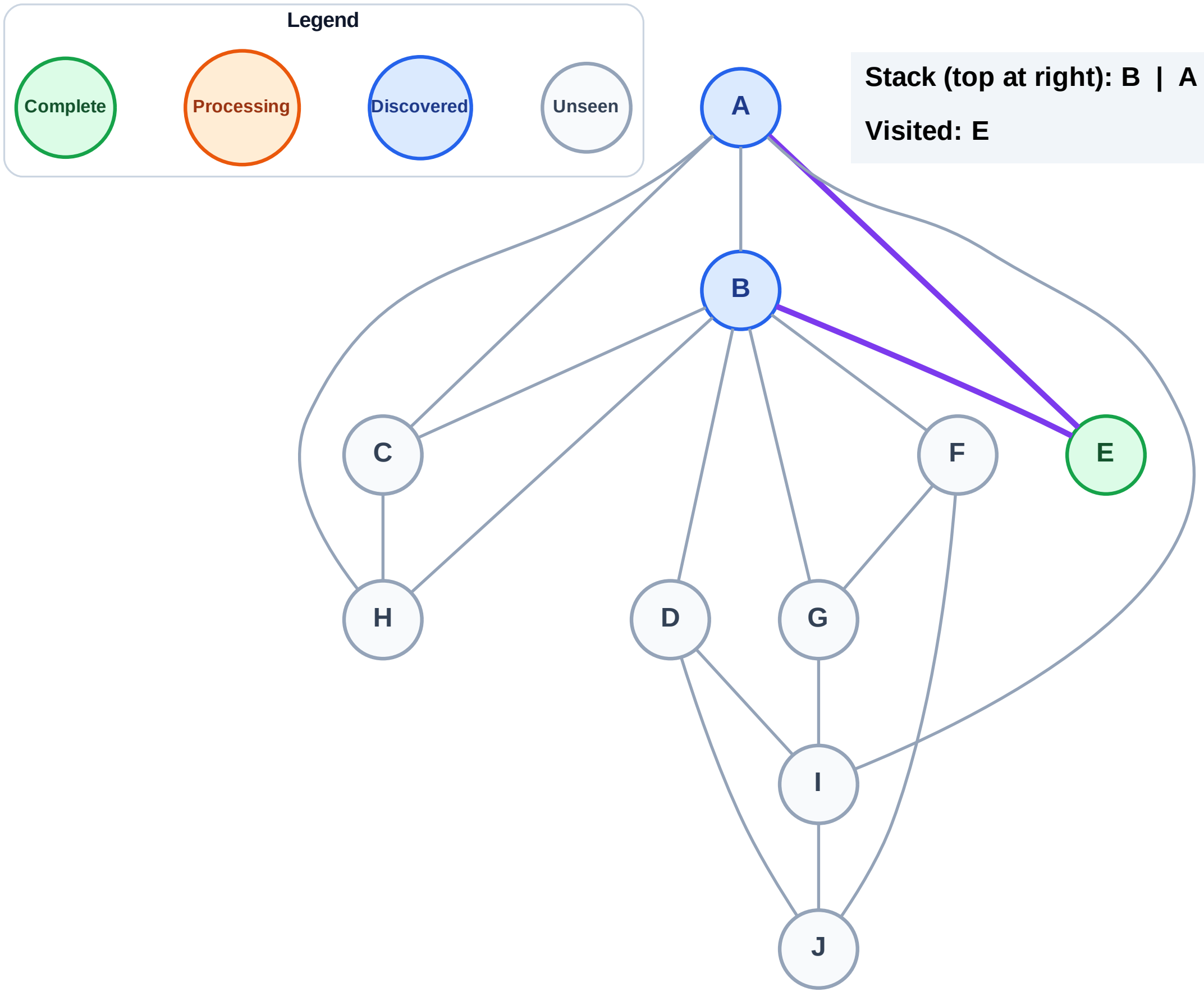
DFS Traversal

Step 2: Process node E



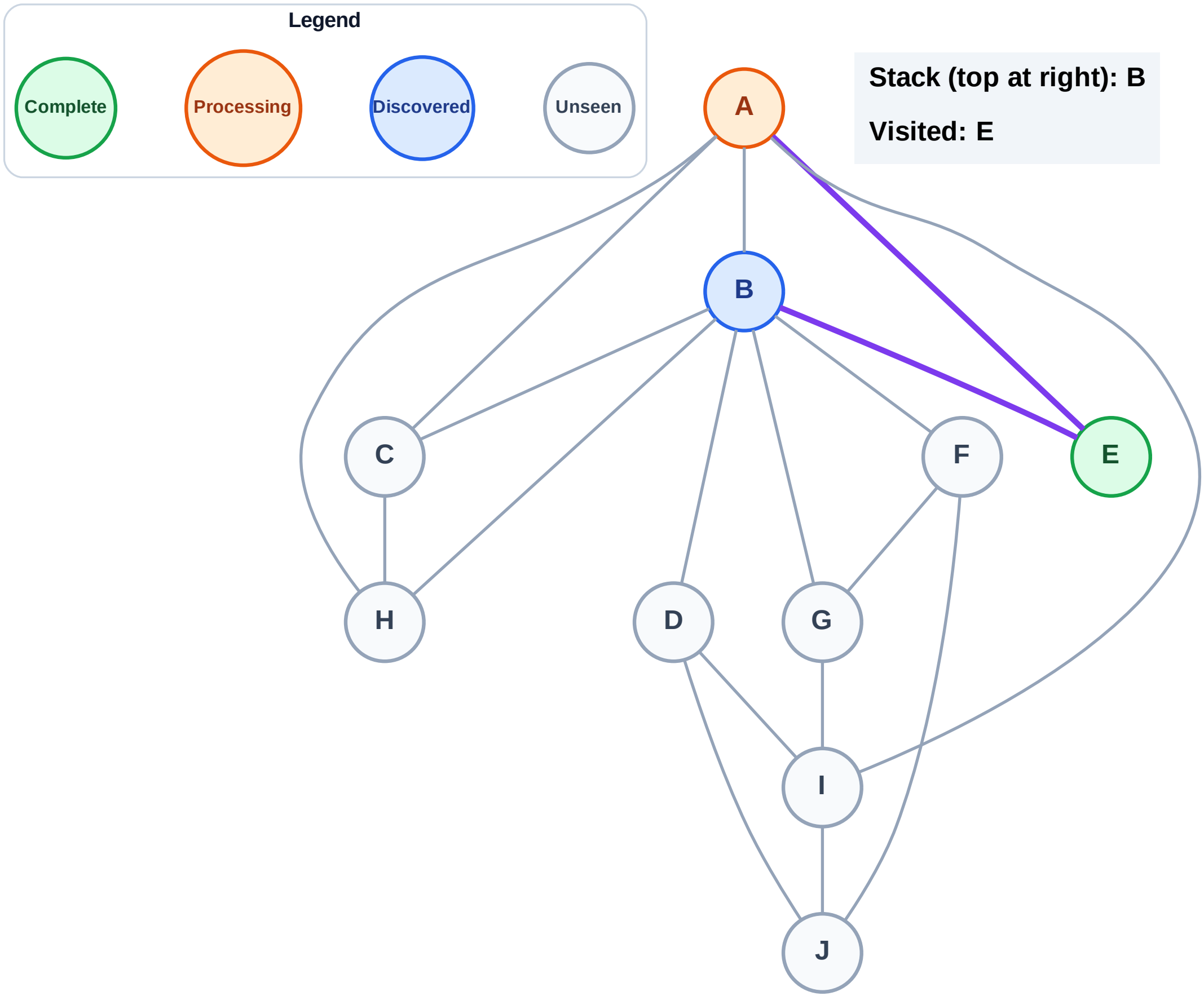
DFS Traversal

Step 3: Discover B, A from E



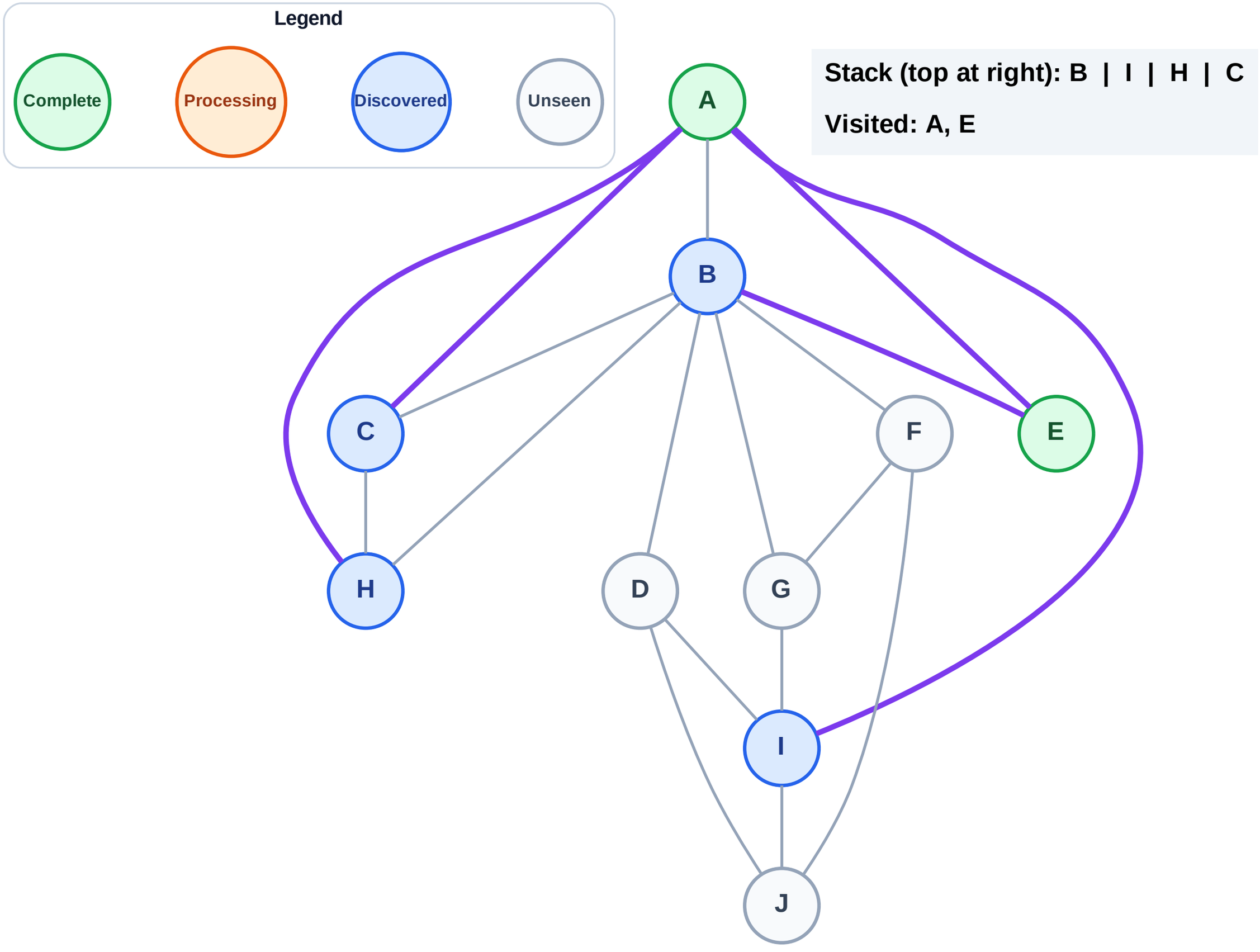
DFS Traversal

Step 4: Process node A



DFS Traversal

Step 5: Discover I, H, C from A

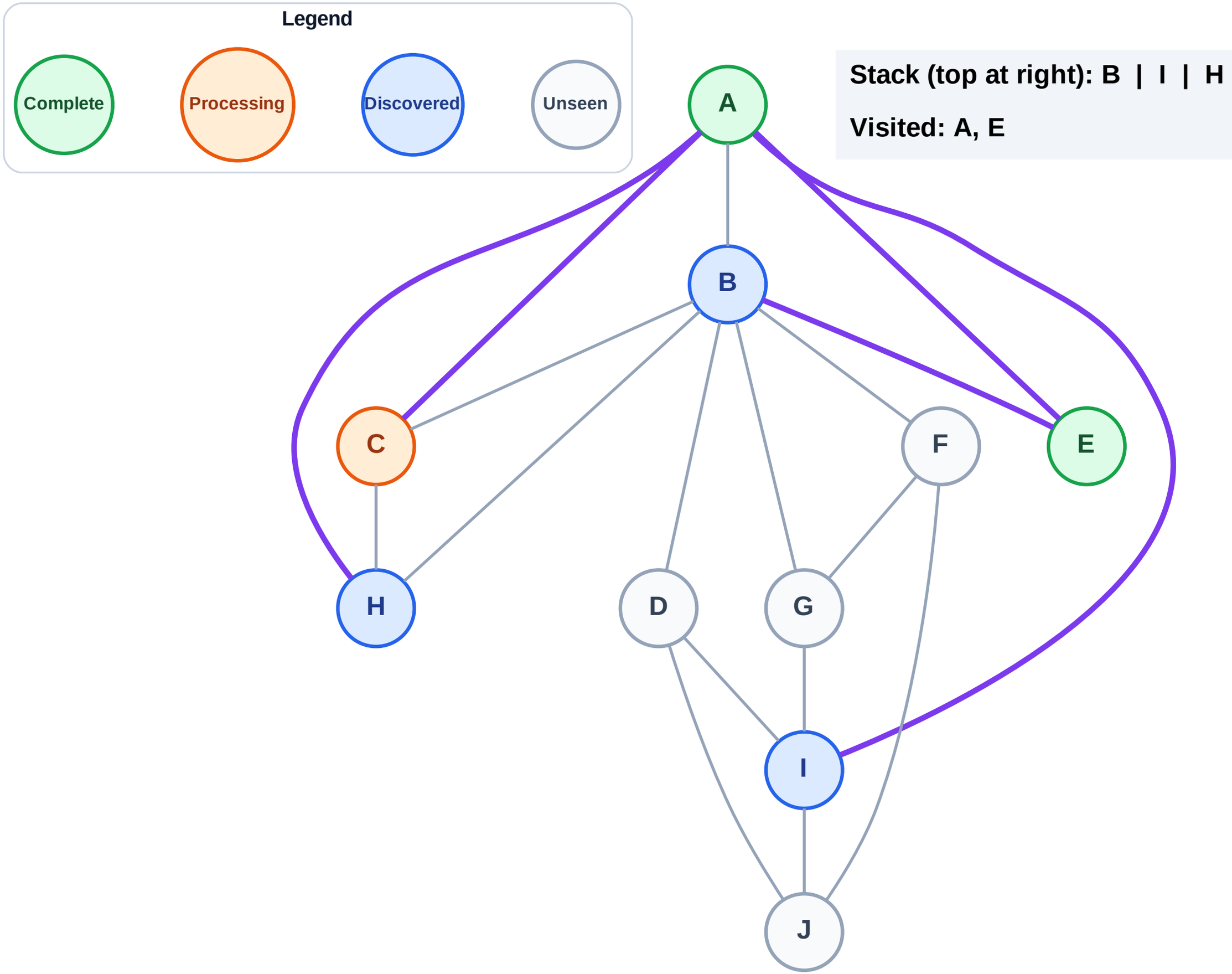


Stack (top at right): B | I | H | C

Visited: A, E

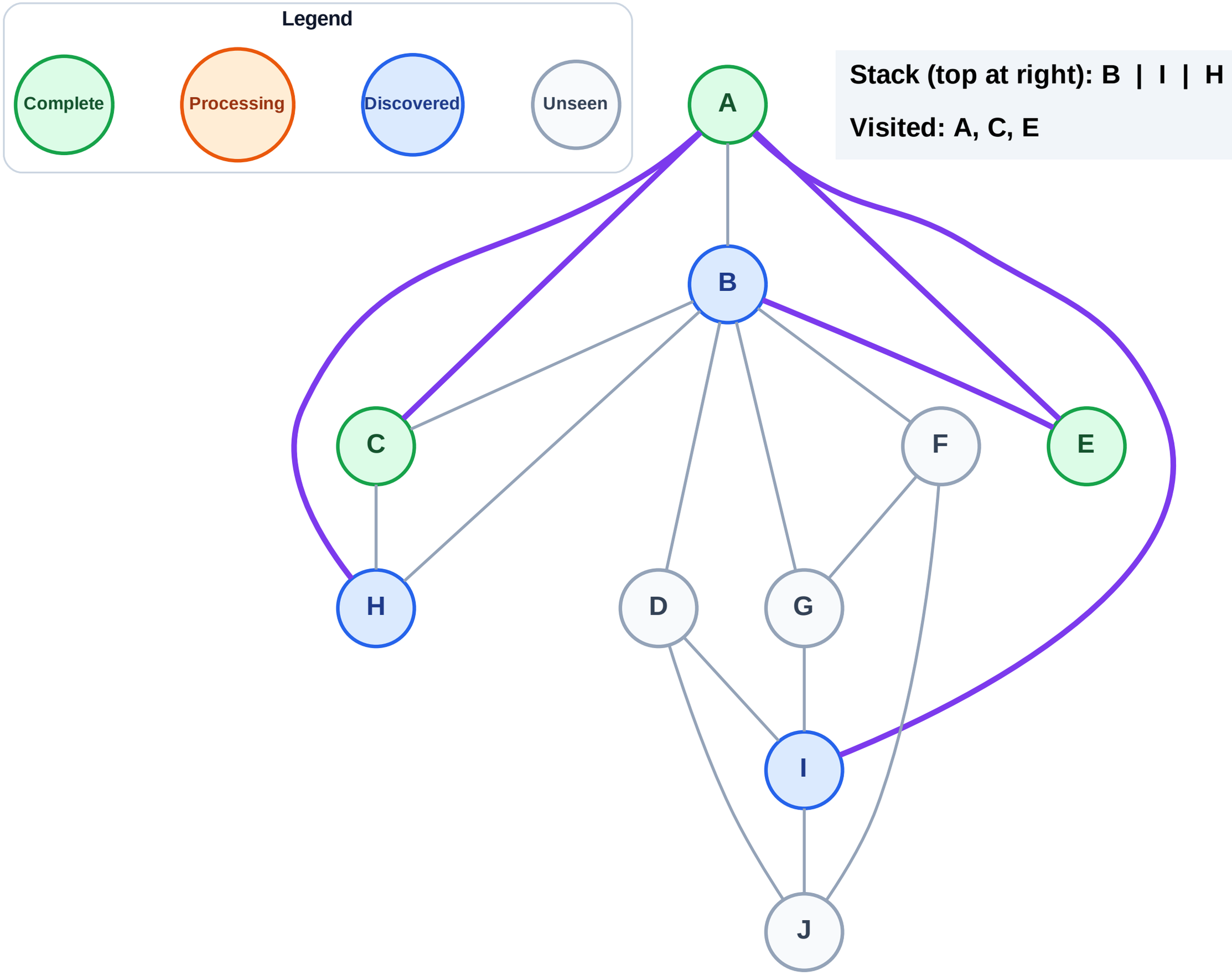
DFS Traversal

Step 6: Process node C



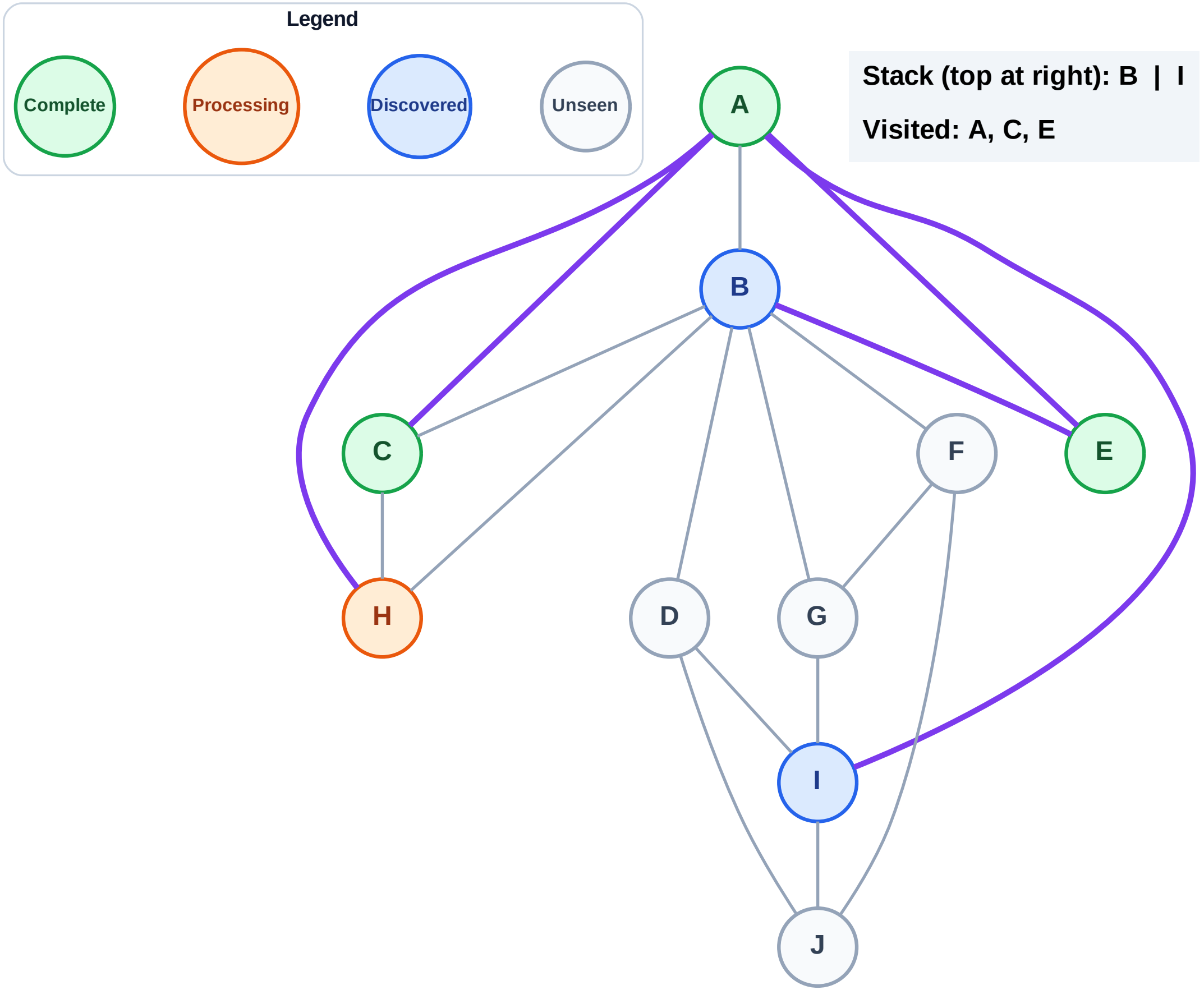
DFS Traversal

Step 7: No new neighbors from C



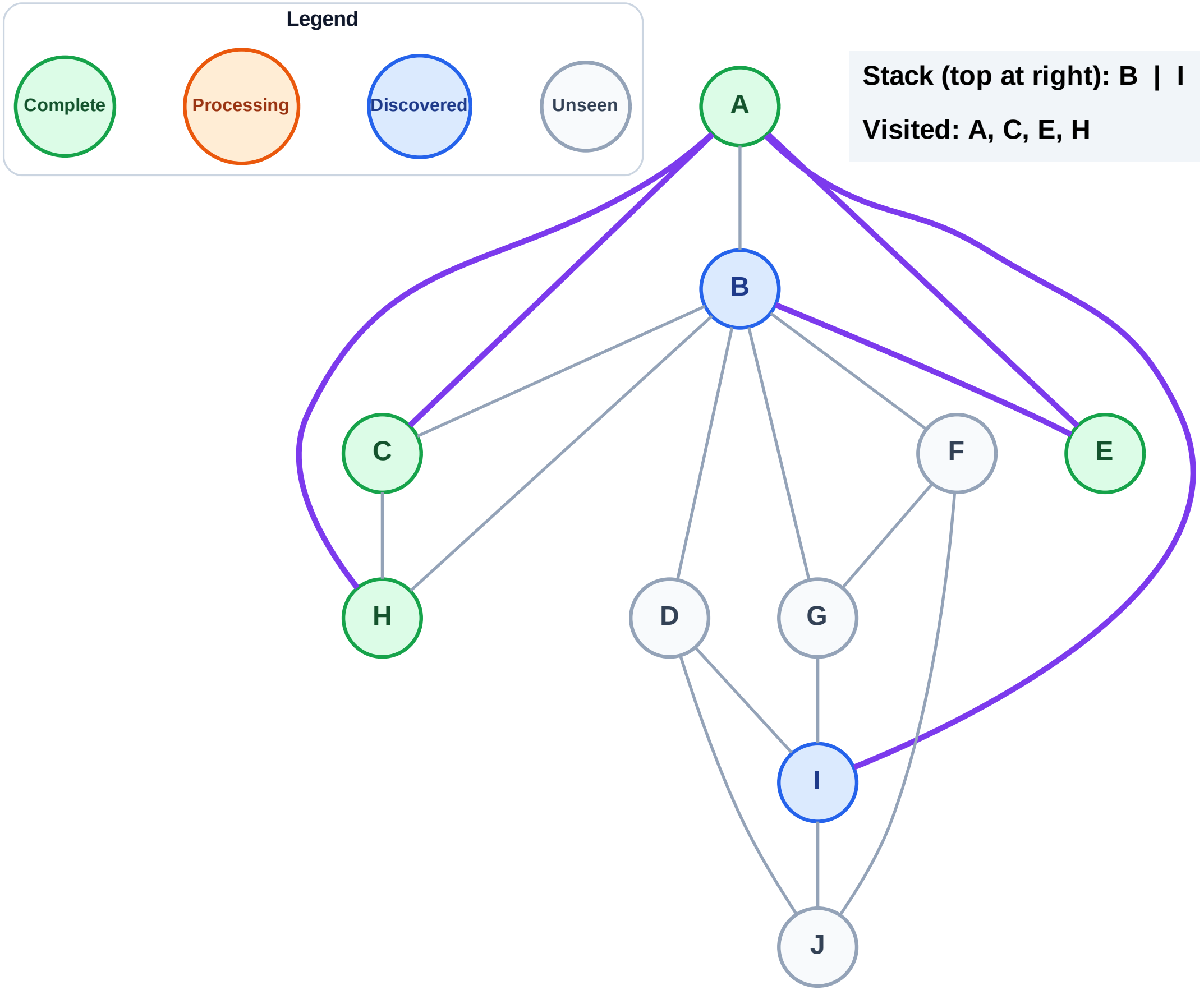
DFS Traversal

Step 8: Process node H



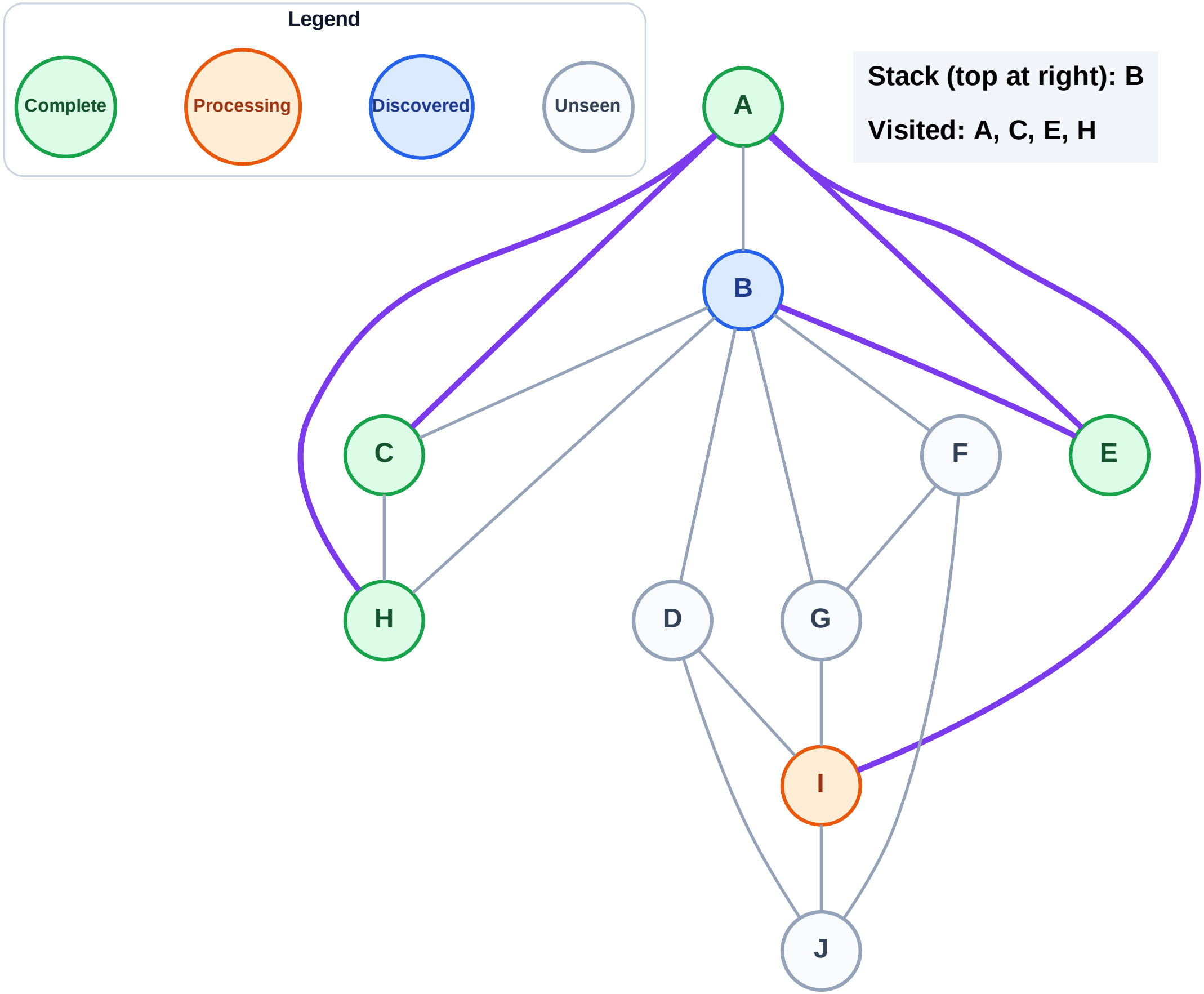
DFS Traversal

Step 9: No new neighbors from H



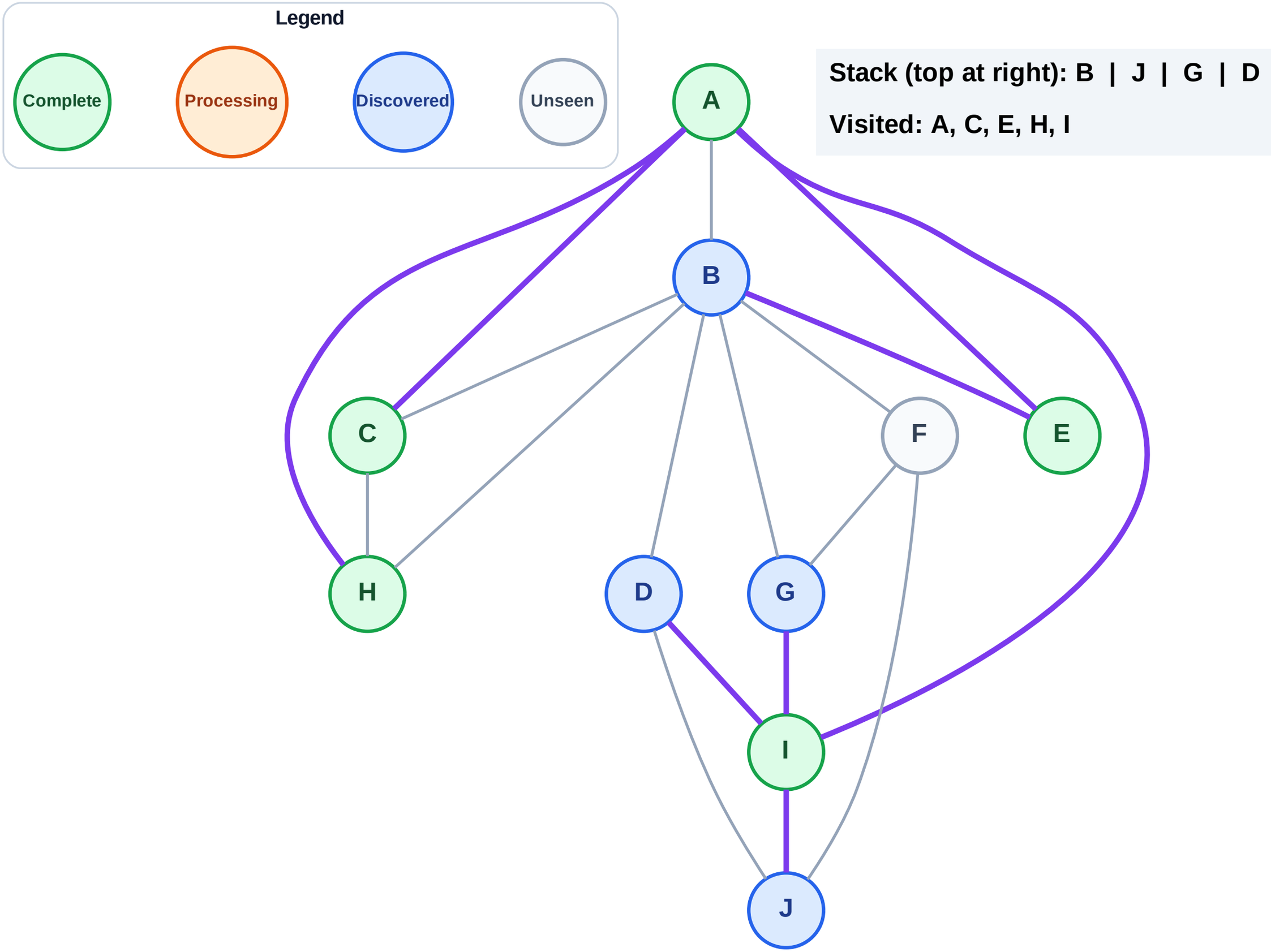
DFS Traversal

Step 10: Process node I



DFS Traversal

Step 11: Discover J, G, D from I

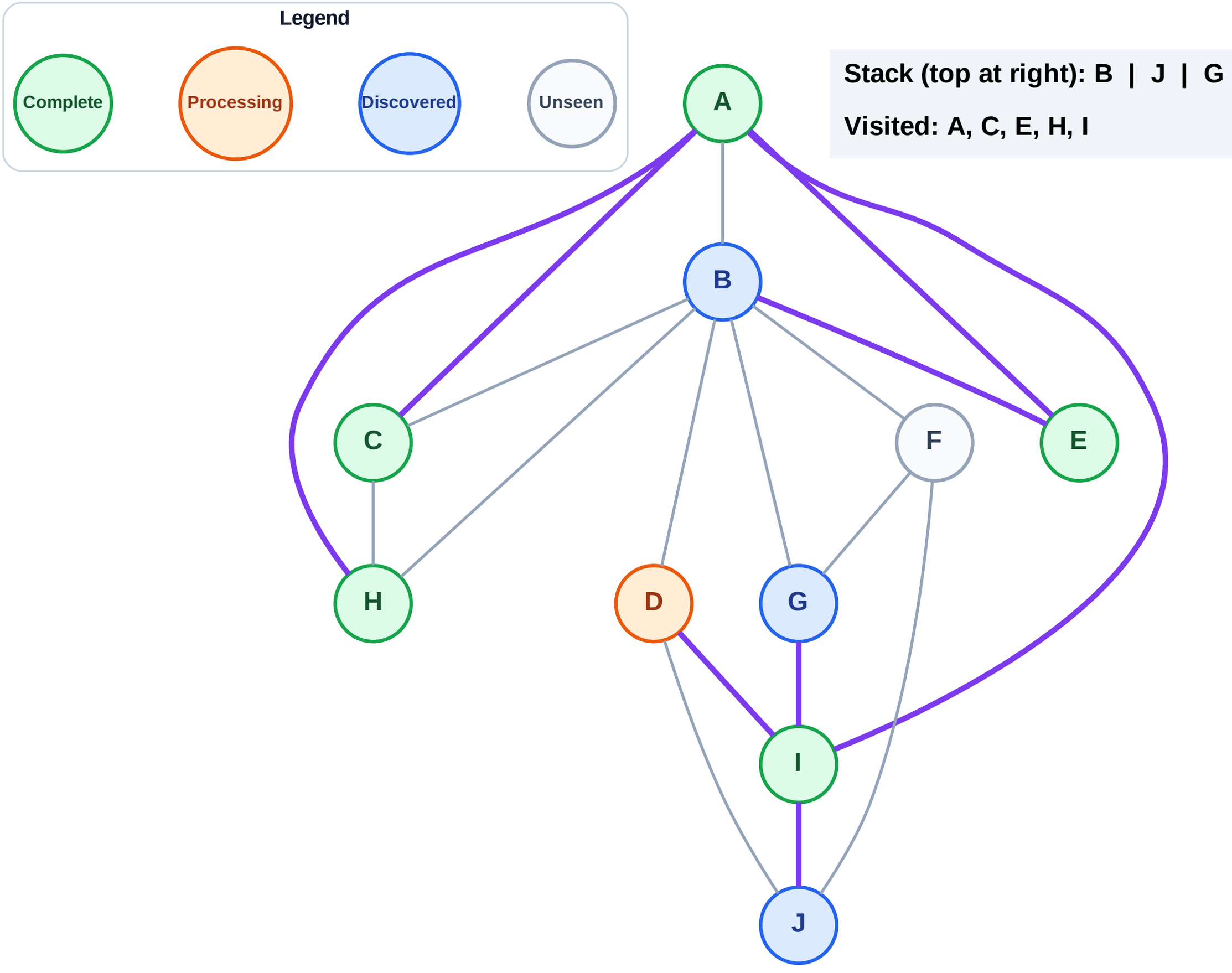


Stack (top at right): B | J | G | D

Visited: A, C, E, H, I

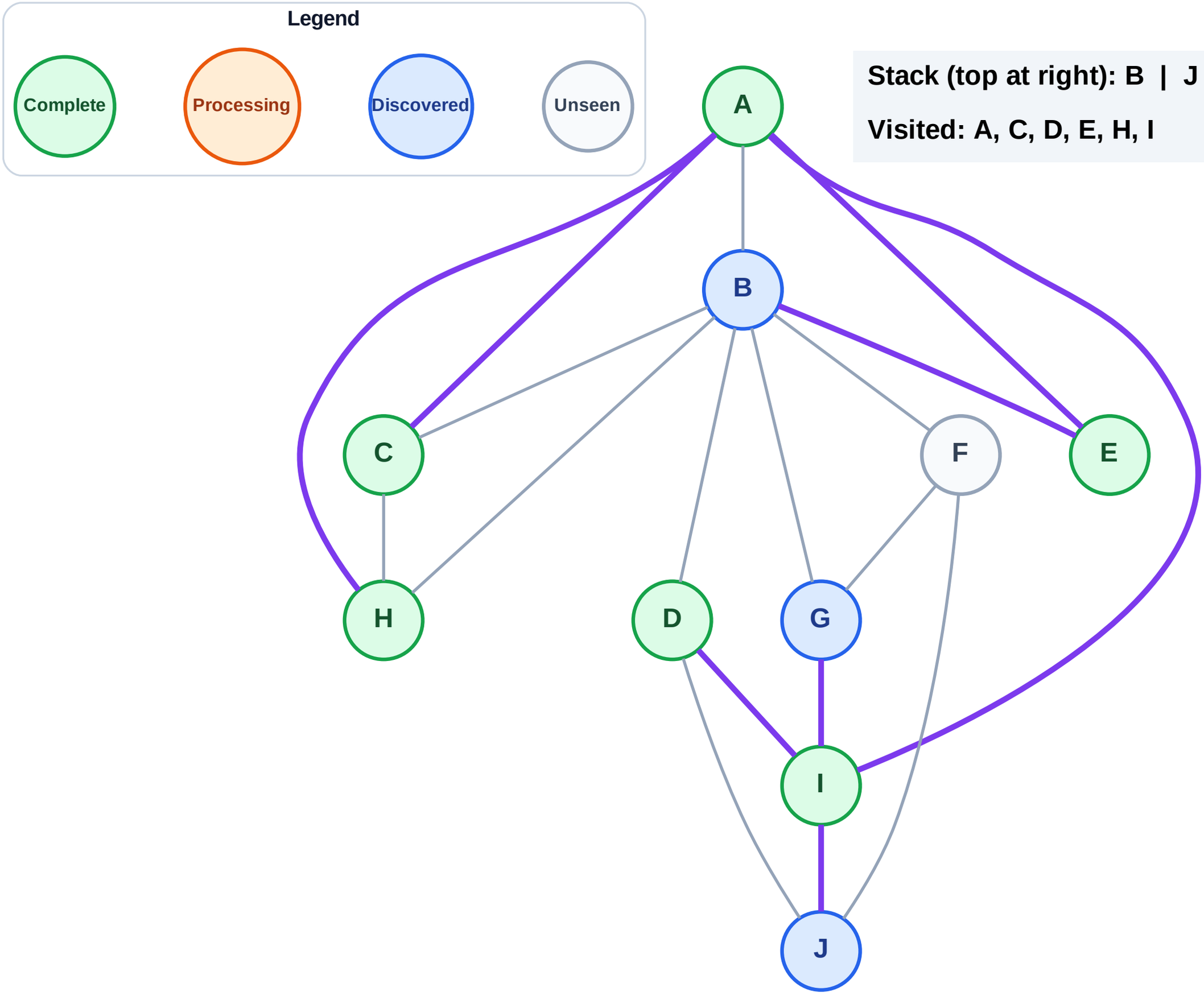
DFS Traversal

Step 12: Process node D



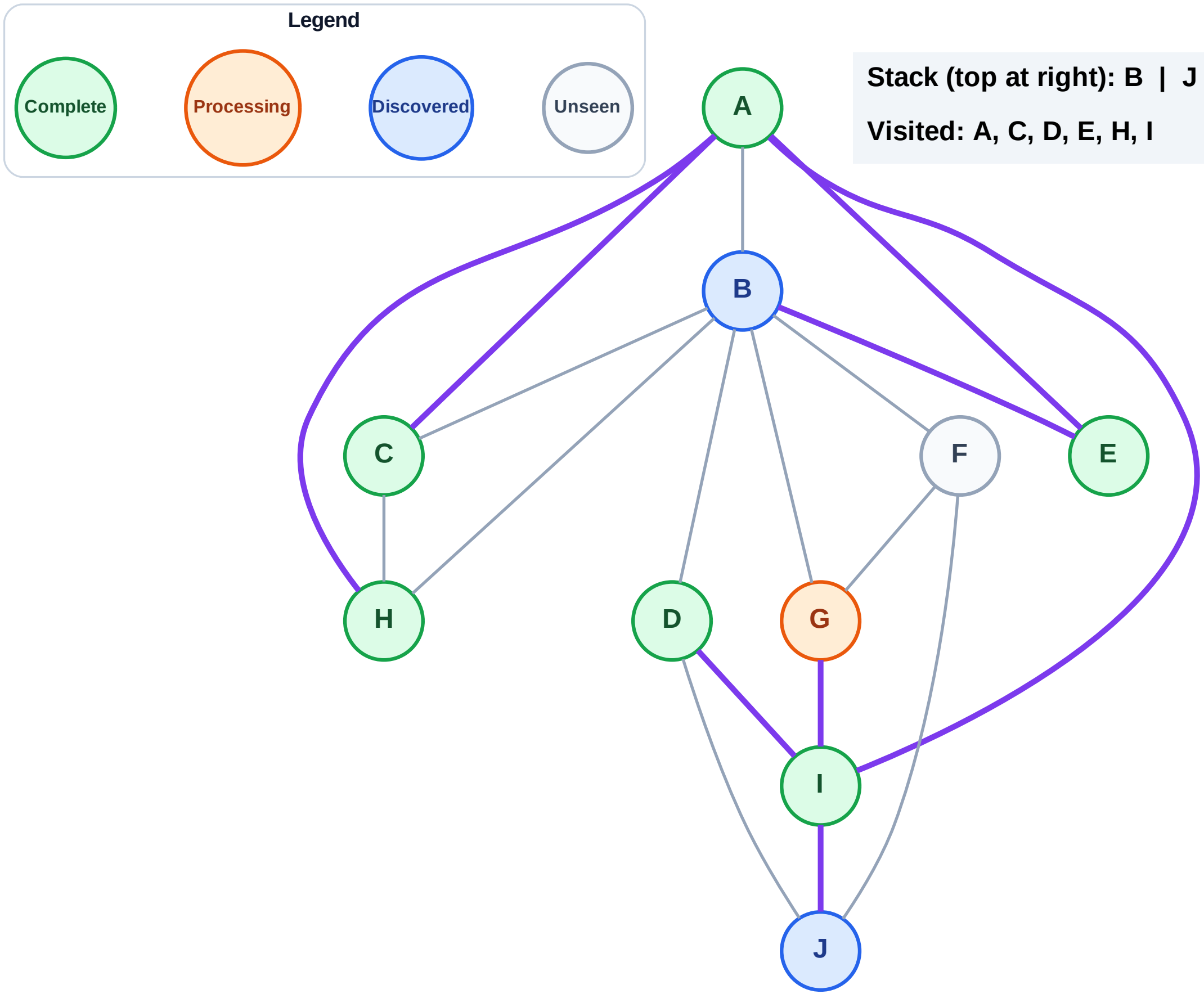
DFS Traversal

Step 13: No new neighbors from D



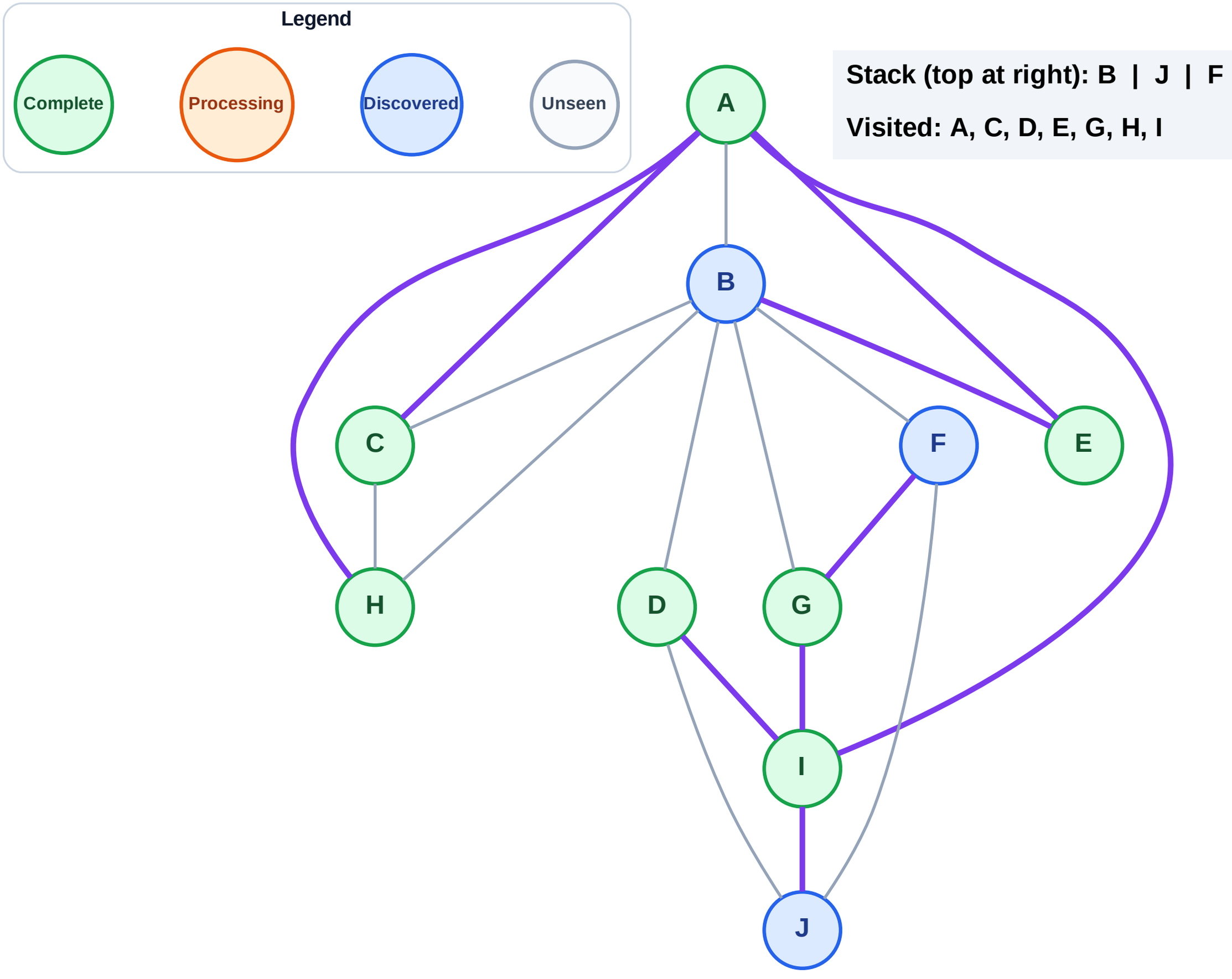
DFS Traversal

Step 14: Process node G



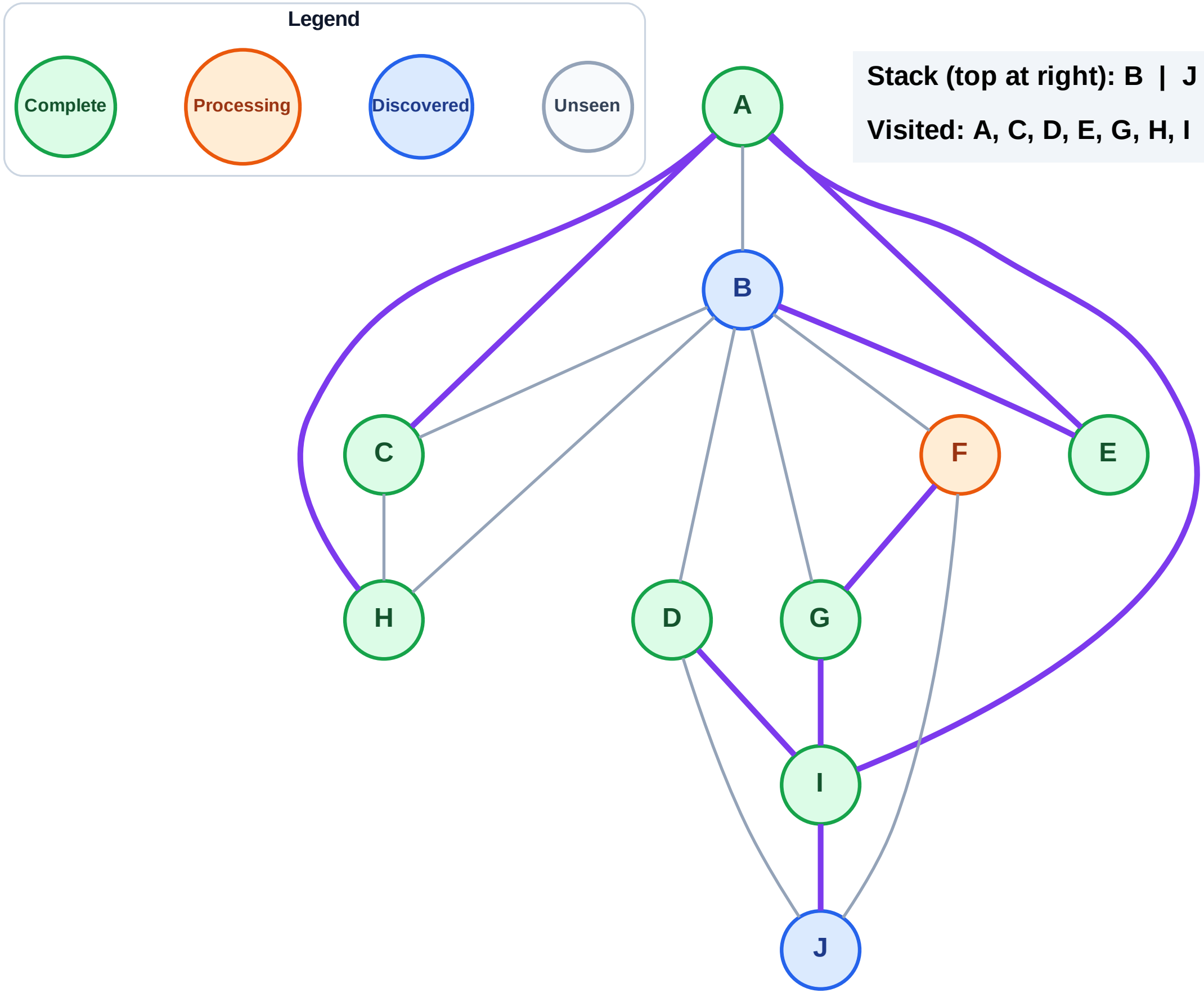
DFS Traversal

Step 15: Discover F from G



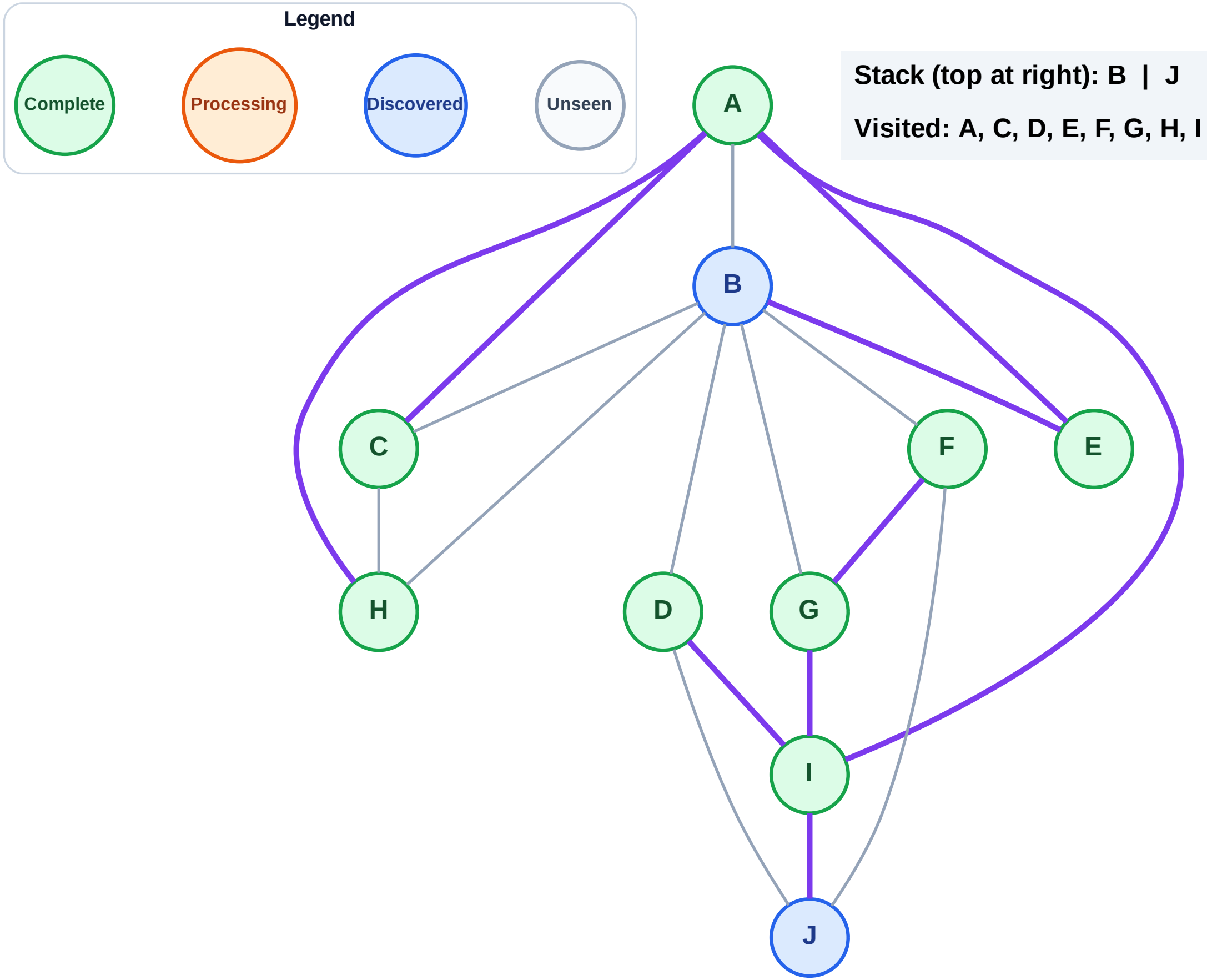
DFS Traversal

Step 16: Process node F



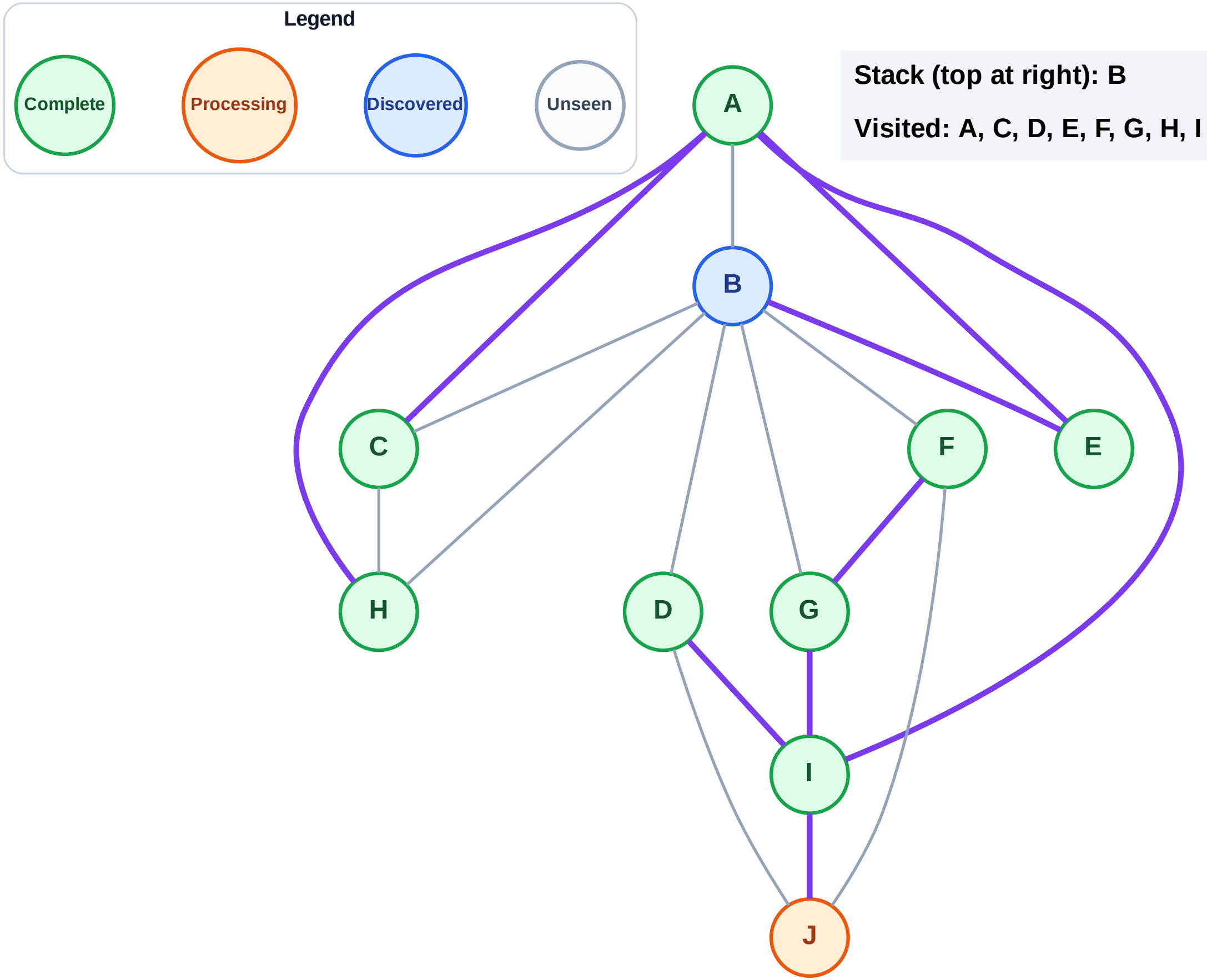
DFS Traversal

Step 17: No new neighbors from F



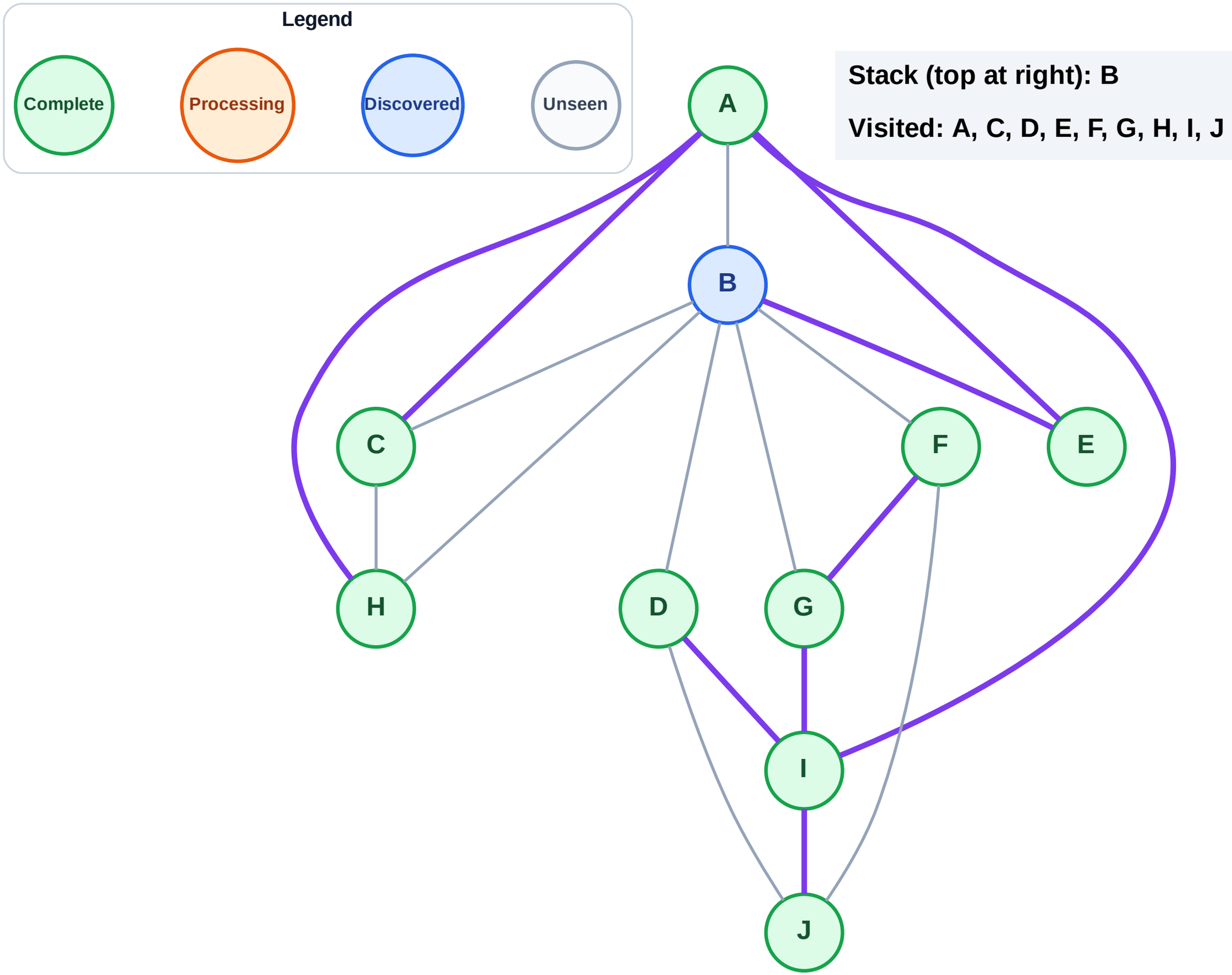
DFS Traversal

Step 18: Process node J



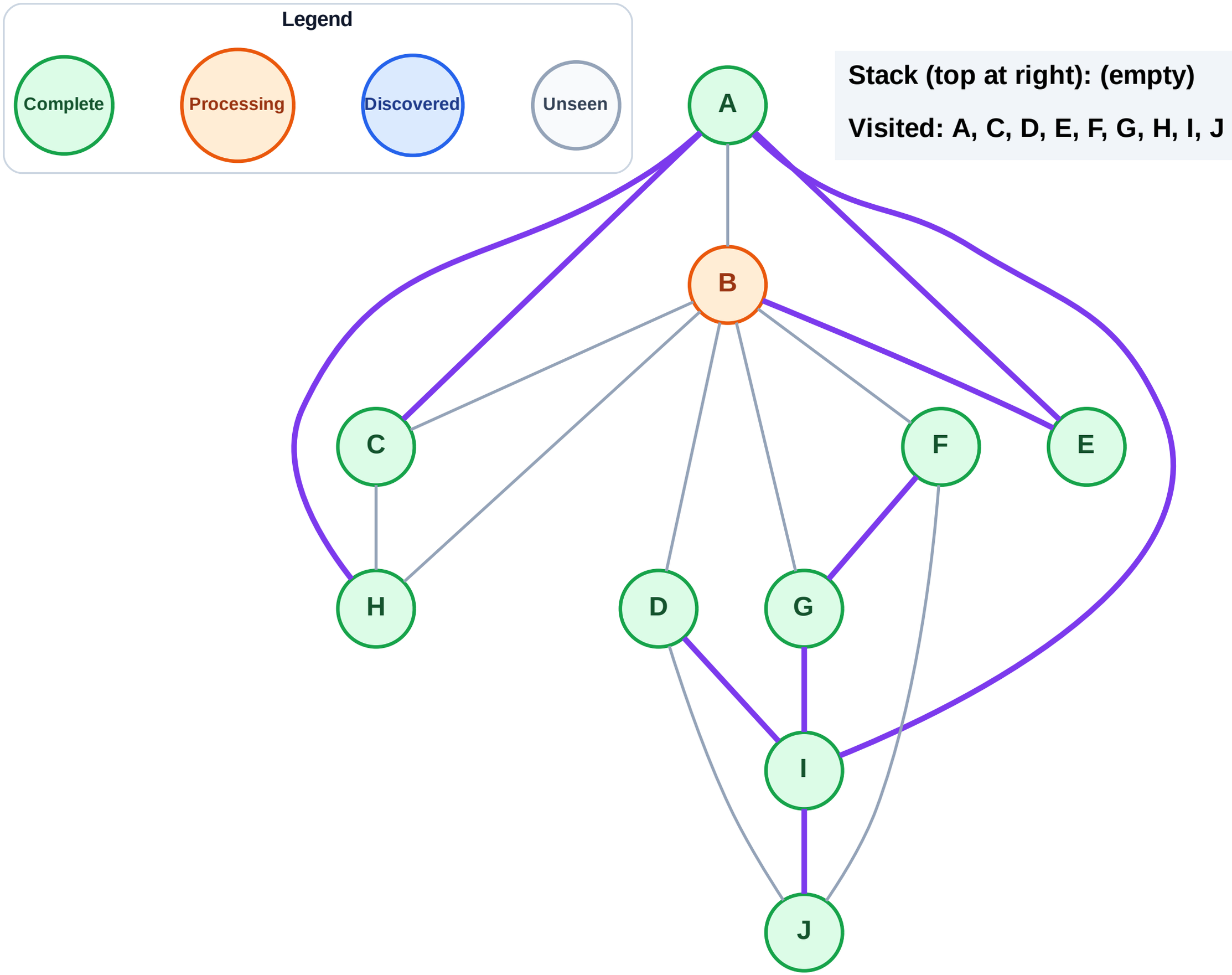
DFS Traversal

Step 19: No new neighbors from J



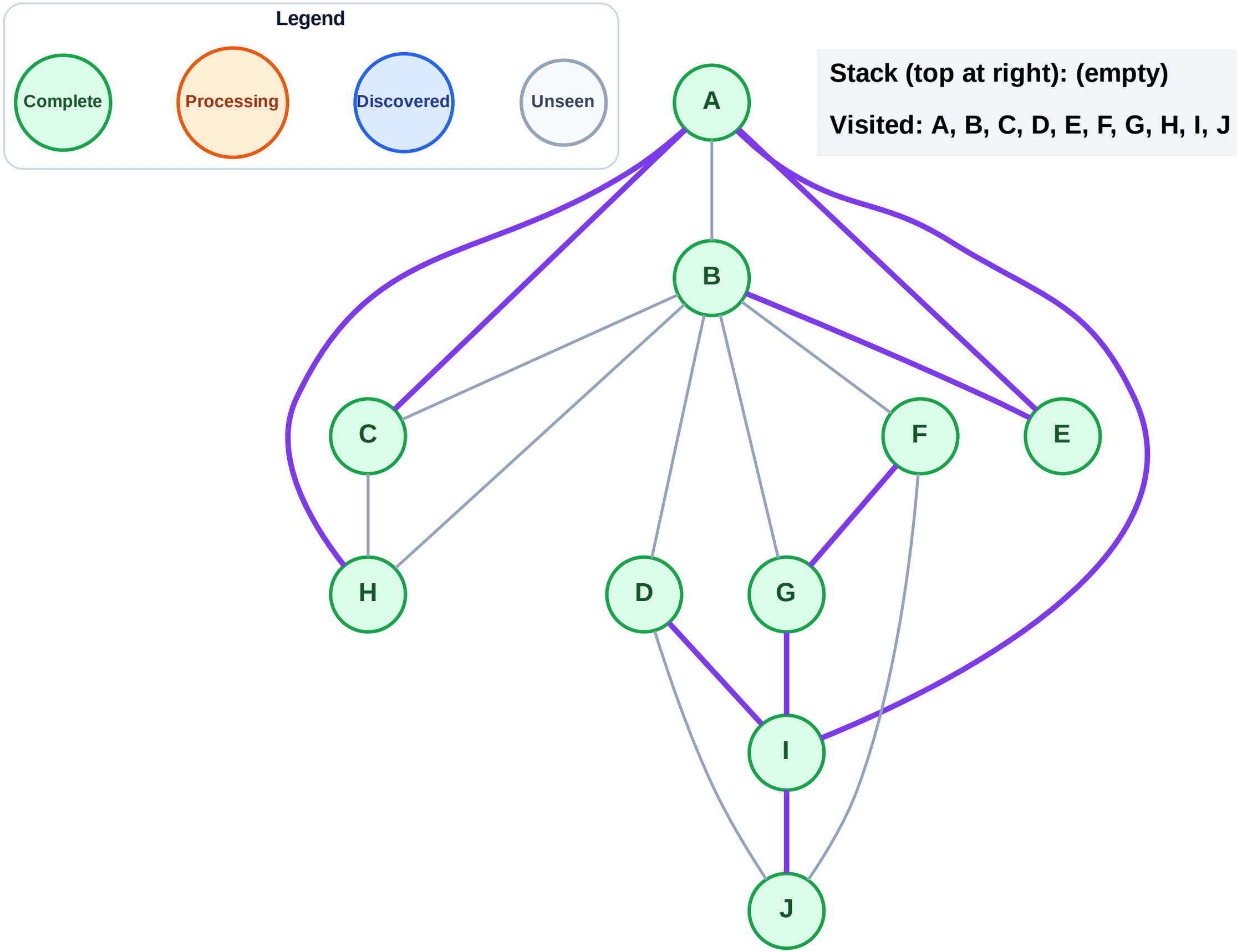
DFS Traversal

Step 20: Process node B



DFS Traversal

Step 21: No new neighbors from B



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J